

Resonant Lerch Limits and Shifted Gamma Correlations

Uniform Herglotz asymptotics and nonseparable distribution jets

ProveIt research continuation

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Abstract

This continuation develops four connections among polylogarithms, Gamma and polygamma functions, Hurwitz and Riemann zeta functions, and generalized Stieltjes constants. First, a centered normalization of the Lerch expansion gives an all-orders resonance theorem uniform in every positive integer order, including orders tending to infinity. Its transition scale is $\log(1/t) + H_k - \gamma$, its leading profile is $(1 - e^{-u})/u$, and its coefficients separate the Stieltjes shift data from polygamma order data. Second, shifted Hurwitz correlations yield exact formulas for the periodic log-Gamma autocorrelation. A convergent odd-zeta expansion proves strict convexity of that autocorrelation and gives its unique minimum, together with an all-order description of the logarithmic cusp at a vanishing shift. Third, a gamma-mixture formula resolves the Herglotz regime in which argument and derivative order grow proportionally. The resulting profile has an explicit finite-order remainder certificate, a divisor partial-fraction expansion, interlacing imaginary zeros after centering, an exponentially small Bessel representation, and normalized Gamma and dilogarithm primitives. Finally, a Frobenius-duality argument extends the formal reflection dimension law to arbitrary active ideals with at most two generators. A concrete three-generator example proves that the unrestricted colength extrapolation fails. Detailed proofs, exact finite-field replay code, numerical diagnostics, source provenance, and integration recommendations accompany the article. Classical input identities are credited, and no numerical relation is promoted to a theorem.

Keywords. Lerch transcendent; polylogarithm order derivatives; log-Gamma correlation; Stieltjes constants; Herglotz function; uniform asymptotics; distribution relations; Koszul homology.

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1 Scope, notation, and the status of the results

This article continues the collective manuscript *Polylogarithms and their Arithmetic Bridges* in Vladimir Reshetnikov’s ProveIt repository. The source review is frozen at commit

570b0567f311cf1890865065896be2665f469e4f.

The two entry points supplied for this task were the canonical manuscript directory and `docs/incoming`. At the frozen revision the latter contains its intake ledger; the relevant previous arrivals have been placed in the thematic reports tree. The canonical README reports a 379-page manuscript and explicitly distinguishes preserved reports, finite replay, and integrated proofs [1]. We follow that distinction.

1.1 What was already available

The reviewed source includes the established S_4 proof, rational distribution-rank theorems, signed-kernel results, log-Gamma moment asymptotics, Stieltjes parameter derivatives, and the existing sharp fixed-argument Herglotz expansion. These form the baseline, rather than new claims of this continuation. In particular, the exact evaluation of the manuscript’s S_6 remains a separate problem. Nothing in the present work proves its special-value conjecture.

Two previous developments are particularly close to the present analysis. The boundary continuation already pairs the Gamma and zeta poles in the Lerch expansion and proves endpoint differentiability thresholds [2]. The integrated zeta-jet chapters already identify Stieltjes primitives with spectral derivatives of Hurwitz zeta at zero [3]. Our first two analytic developments connect and extend those mechanisms in different directions: uniformity in a growing integer order, and nonzero shifts in bilinear integrals.

1.2 The principal additions

Table 1 separates the new conclusions proved here from their classical or previously established inputs. “New” in this table means a continuation beyond the inspected ProveIt material; no claim of worldwide priority is made without a separate literature investigation.

Table 1: Contributions and their established inputs.

Topic	Conclusion established here	Input credited separately
Resonant Lerch limits	An arbitrary-order expansion uniform in all integer $k \geq 0$, with a common centered scale and an exponentially small analytic tail	Classical Lerch expansion; fixed-order pole cancellation
Shifted Gamma correlations	Exact autocorrelation and increment formulas; strict convexity of the autocorrelation; unique half-period extremum; mixed Hurwitz-jet cusps	Hurwitz Fourier expansion; classical unshifted integral calculus
Herglotz joint limit	Complete compact-ratio expansion and a finite- r Taylor bound; a profile with arithmetic, Bessel, Gamma, and dilogarithm representations	Digamma integral; classical partial fractions and zeta functional equation
Nonseparable jets	A two-generator colength theorem in any local Frobenius algebra; exact reflection dimensions and integer torsion; three-generator obstruction	The manuscript’s all-ring distribution basis and reflection–Koszul model

The analytic results have complementary roles. The Lerch theorem controls a moving spectral order near an integer. The Gamma correlation theorem integrates two spectral jets against a periodic shift. The Herglotz theorem controls a moving argument when differentiation order grows. The final algebraic section concerns the formal distribution relations satisfied by families of these functions; it makes no assertion of transcendental independence of their evaluated values.

1.3 Conventions

We use the Hurwitz Laurent convention

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n=0}^{\infty} \frac{(-1)^n \gamma_n(a)}{n!} (s-1)^n. \quad (1.1)$$

Thus $\gamma_0(a) = -\psi(a)$ and $\gamma_0(1) = \gamma$, Euler's constant. Square-bracket superscripts denote spectral derivatives:

$$\zeta^{[r]}(s, a) = \partial_s^r \zeta(s, a), \quad \text{Li}_s^{[r]}(z) = \partial_s^r \text{Li}_s(z).$$

An ordinary prime on $\zeta(s)$ also denotes an s derivative. Parenthesized superscripts on ψ , Γ , or a named real-variable function denote derivatives in that function's argument. Each section specifies any further local notation. All logarithms of positive real numbers are real logarithms; complex logarithms and square roots are principal unless another branch is explicitly specified. The fractional part $\{x\}$ is used only under integrals or away from its integer discontinuities, so values assigned at the isolated endpoints do not affect the integrals.

The source identities for Hurwitz zeta, Gamma, and Lerch are recorded in DLMF [6, 7]. In particular,

$$\zeta^{[1]}(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi), \quad \partial_a \zeta(s, a) = -s \zeta(s+1, a). \quad (1.2)$$

These identities fix the normalization of every antiderivative below.

1.4 How the evidence is used

Theorems in this article are proved analytically or by exact linear algebra. The computational artifacts have three distinct purposes. Symbolic recurrences and finite-field ranks are checked exactly. Independent high-precision evaluations test normalizations and signs in the analytic identities. Asymptotic tables illustrate the stated remainder orders, without claiming that a finite table proves a uniform estimate. None of the proofs depends on accepting a small floating-point residual as zero.

2 A resonance theorem uniform in the integer order

The Lerch expansion has two singular summands when the order approaches a positive integer. Their sum is regular. Classical formulas describe this cancellation for a fixed integer order; the useful additional point here is a normalization that controls *every* integer order at once. It allows the order, its distance from an integer, and the distance of the argument from one to vary simultaneously.

2.1 The classical identity and the normalized remainder

For $a > 0$ and $t > 0$ put

$$F(s, t, a) = e^{-at} \Phi(e^{-t}, s, a) = \sum_{n=0}^{\infty} \frac{e^{-(n+a)t}}{(n+a)^s}. \quad (2.1)$$

In particular $F(s, t, 1) = \text{Li}_s(e^{-t})$. The classical Lerch expansion is

$$F(s, t, a) = \Gamma(1-s)t^{s-1} + \sum_{j=0}^{\infty} \zeta(s-j, a) \frac{(-t)^j}{j!}, \quad 0 < t < 2\pi, \quad (2.2)$$

initially for $s \notin \{1, 2, \dots\}$. We use the locally normally convergent form of this identity; see DLMF 25.14.3_1 [7]. Integer-order polylogarithm derivatives obtained by pairing the poles are already treated by Bailey and Borwein [8]. The prior boundary continuation in ProveIt also uses this cancellation to establish sharp endpoint regularity [2]. We do not claim those facts as new.

Let $k \geq 0$ be an integer and write $s = k + 1 + \varepsilon$. Empty sums and products have their usual values. Define

$$\mathcal{R}_k(\varepsilon, t, a) = \frac{k!}{(-t)^k} \left[F(k+1+\varepsilon, t, a) - \sum_{j=0}^{k-1} \zeta(k+1+\varepsilon-j, a) \frac{(-t)^j}{j!} \right]. \quad (2.3)$$

This subtraction removes the regular terms that precede the resonance. It is an exact definition, even when numerical evaluation by direct subtraction would lose many digits. Introduce

$$H_k^{(r)} = \sum_{j=1}^k j^{-r}, \quad H_k = H_k^{(1)}, \quad (2.4)$$

$$T = \log(1/t) + H_k - \gamma, \quad (2.5)$$

$$G_a(\varepsilon) = \zeta(1+\varepsilon, a) - \frac{1}{\varepsilon} = \sum_{r=0}^{\infty} \frac{(-1)^r \gamma_r(a)}{r!} \varepsilon^r.$$

The apparent singularity of G_a is removable. Throughout the theorem below T is positive and tends to infinity. No separate upper bound on k is imposed.

Definition 2.1 (Centered resonance coefficients). For $|w| < 1$, define

$$E_k(w) = \exp \left\{ \sum_{r=2}^{\infty} \frac{\zeta(r) + (-1)^r H_k^{(r)}}{r} w^r \right\} = \sum_{j=0}^{\infty} b_{k,j} w^j. \quad (2.6)$$

Thus $b_{k,0} = 1$ and $b_{k,1} = 0$. If $c_{k,r} = \zeta(r) + (-1)^r H_k^{(r)}$, the coefficients can be generated by

$$j b_{k,j} = \sum_{r=2}^j c_{k,r} b_{k,j-r} \quad (j \geq 2). \quad (2.7)$$

These coefficients are polygamma polynomials. More explicitly,

$$c_{k,r} = (1 + (-1)^r) \zeta(r) - \frac{\psi^{(r-1)}(k+1)}{(r-1)!}, \quad r \geq 2. \quad (2.8)$$

Equation (2.8) follows directly from the polygamma series. It also gives the complete Bell-polynomial description of $b_{k,j}$, with cumulants $(r-1)! c_{k,r}$ and zero first cumulant.

Lemma 2.2 (Exact cancellation). For $|\varepsilon| < 1/2$ and $0 < t < 2\pi$,

$$\mathcal{R}_k(\varepsilon, t, a) = \frac{1 - e^{-T\varepsilon} E_k(\varepsilon)}{\varepsilon} + G_a(\varepsilon) + B_k(\varepsilon, t, a), \quad (2.9)$$

$$B_k(\varepsilon, t, a) = \sum_{\ell=1}^{\infty} \frac{k! (-t)^\ell}{(k+\ell)!} \zeta(1+\varepsilon-\ell, a). \quad (2.10)$$

The quotient at $\varepsilon = 0$ is interpreted by continuity.

Proof. The $j = k$ term of (2.2) becomes $\zeta(1 + \varepsilon, a)$ after normalization. The gamma recurrence gives

$$\frac{k!t^{k+\varepsilon}\Gamma(-k-\varepsilon)}{(-t)^k} = -\frac{t^\varepsilon\Gamma(1-\varepsilon)}{\varepsilon\prod_{j=1}^k(1+\varepsilon/j)}.$$

The Taylor series of $\log[\Gamma(1-\varepsilon)/\prod_{j=1}^k(1+\varepsilon/j)]$ is

$$(\gamma - H_k)\varepsilon + \sum_{r=2}^{\infty} \frac{c_{k,r}}{r} \varepsilon^r.$$

This proves (2.9); the remaining terms in the normally convergent series are exactly (2.10). Pairing the two poles first makes the identity holomorphic at zero, so no value is assigned to either divergent summand separately. $\square$

Theorem 2.3 (Uniform resonance, including growing integer order). *Fix $0 < t_0 < 2\pi$, a compact set $A \subset (0, \infty)$, a compact set $U \subset \mathbb{C}$, and integers $M, J \geq 0$. Let $k \geq 0$, $0 < t \leq t_0$, $a \in A$, $u \in U$, and T be as in (2.4). As $T \rightarrow \infty$, uniformly over all these choices,*

$$\begin{aligned} \mathcal{R}_k(u/T, t, a) &= Th(u) + \gamma_0(a) \\ &\quad + \sum_{r=1}^M \frac{u^r}{T^r} \left\{ \frac{(-1)^r \gamma_r(a)}{r!} - e^{-u} b_{k,r+1} \right\} + O(T^{-M-1}), \end{aligned} \quad (2.11)$$

$$h(u) = \frac{1 - e^{-u}}{u} = \int_0^1 e^{-uv} dv, \quad h(0) = 1. \quad (2.12)$$

The error and its first J derivatives in u are $O(T^{-M-1})$. The same claim holds after any fixed number of derivatives in a on A . Before absorbing the analytic tail into the error, it has the stronger bound

$$B_k(u/T, t, a) = O\left(\frac{t}{k+1}\right) = O(e^{-T}). \quad (2.13)$$

All constants may depend on the displayed compact sets and derivative orders, but are independent of k and t .

Proof. We give the uniformity argument, since it is what permits the integer order to grow. Choose $0 < \eta < 1/2$. Since $|c_{k,r}| \leq 2\zeta(r)$ for every k and $r \geq 2$, the series

$$2 \sum_{r=2}^{\infty} \frac{\zeta(r)}{r} \eta^r$$

is a common upper bound for $|\log E_k(w)|$ on $|w| \leq \eta$. Consequently the functions E_k have Taylor remainders bounded uniformly in k on every smaller disk. The functions G_a have the same property, uniformly for a in a compact complex neighborhood of A contained in $\operatorname{Re} a > 0$.

It remains to control the infinite analytic tail uniformly in k . For $\ell \geq 1$,

$$\frac{k!}{(k+\ell)!} \leq \frac{1}{(k+1)\ell!}.$$

Choose $t_0 < R < 2\pi$. Normal convergence of the regular part of the Lerch series, uniformly for $|\varepsilon| \leq \eta$ and a in the chosen compact neighborhood, gives a finite common bound for

$$\sum_{\ell=1}^{\infty} \frac{t_0^{\ell-1}}{\ell!} |\zeta(1 + \varepsilon - \ell, a)|.$$

For an explicit justification of this parameter-uniform normal convergence, use Hermite's Hurwitz integral [6, 25.11.29]. Its integral part is

$$i \int_0^\infty \frac{(a + iy)^{-s} - (a - iy)^{-s}}{e^{2\pi y} - 1} dy.$$

At $s = 1 + \varepsilon - \ell$, the sum of absolute majorants weighted by $R^\ell/\ell!$ is bounded for $y \geq 1$ by a fixed polynomial in y times $e^{-(2\pi-R)y}$. For $0 \leq y \leq 1$, the difference of the two powers gains a factor y , which cancels the zero of the denominator; the remaining factorial-weighted series is uniformly bounded. The two elementary terms of Hermite's formula have normally convergent exponential majorants as well. The compact neighborhood has $\operatorname{Re} a$ bounded away from zero, so the argument also applies to complex a and, by Cauchy's formula, to fixed derivatives. This proves the required bound independently of k .

Thus $|B_k| \leq Ct/(k+1)$. The elementary inequality $H_k - \gamma \leq \log(k+1)$ proves the second bound in (2.13). For example, this harmonic inequality follows by comparing the increasing sequence $H_k - \log(k+1)$ with its limit γ .

Now put $\varepsilon = u/T$ in (2.9). The first quotient is

$$Th(u) - e^{-u} \sum_{j=2}^{\infty} b_{k,j} (u/T)^{j-1}.$$

Its apparent singularity at $u = 0$ is removable. Uniform Taylor estimates for this series and for $G_a(u/T)$ give (2.11); the exponentially small tail is smaller than any specified power of T^{-1} . To justify differentiated error estimates, first prove these bounds on slightly larger compact sets in u and a , then apply Cauchy's formula. This avoids differentiating an unspecified asymptotic error. $\square$

Remark 2.4 (What the uniform statement covers). For fixed k , the theorem describes $t \downarrow 0$. It also holds when $k \rightarrow \infty$ with t fixed in $(0, 2\pi)$, or along arbitrary simultaneous sequences for which $T \rightarrow \infty$. The parameter u remains in a fixed compact set. The theorem does not assert the same error for unrestricted $u \rightarrow \infty$, or uniformity as a approaches zero.

2.2 Explicit corrections and the high-order coefficient limit

The first coefficients are

$$\begin{aligned} b_{k,2} &= \frac{1}{2}(\zeta(2) + H_k^{(2)}), \\ b_{k,3} &= \frac{1}{3}(\zeta(3) - H_k^{(3)}), \\ b_{k,4} &= \frac{1}{4}(\zeta(4) + H_k^{(4)}) + \frac{1}{8}(\zeta(2) + H_k^{(2)})^2. \end{aligned} \tag{2.14}$$

In particular,

$$\begin{aligned} \mathcal{R}_k(u/T, t, a) &= Th(u) - \psi(a) - \frac{u}{T} \left[\gamma_1(a) + \frac{e^{-u}}{2}(\zeta(2) + H_k^{(2)}) \right] \\ &\quad + \frac{u^2}{T^2} \left[\frac{\gamma_2(a)}{2} - \frac{e^{-u}}{3}(\zeta(3) - H_k^{(3)}) \right] + O(T^{-3}). \end{aligned} \tag{2.15}$$

Proposition 2.5 (The limiting coefficient function). *Locally uniformly for $|w| < 1$,*

$$E_k(w) \longrightarrow \Gamma(1-w)\Gamma(1+w) = \frac{\pi w}{\sin \pi w}. \tag{2.16}$$

Hence $b_{k,2j+1} \rightarrow 0$ and the even coefficients tend to those of $\pi w/\sin \pi w$. The first nonconstant limits are $b_{k,2} \rightarrow \pi^2/6$ and $b_{k,4} \rightarrow 7\pi^4/360$.

Proof. For every $r \geq 2$, $H_k^{(r)} \rightarrow \zeta(r)$. Dominated convergence in the logarithmic series leaves $2 \sum_{j \geq 1} \zeta(2j) w^{2j} / (2j)$, which is the logarithm of the gamma product in (2.16). The reflection formula gives its sine expression. Local uniform convergence permits coefficient extraction. $\square$

This is a useful separation of roles. The generalized Stieltjes constants carry the shift a ; harmonic numbers and polygamma values carry the integer order; the function h carries the approach to resonance. The scale $T = \log(1/t) + H_k - \gamma$ combines the two sources of logarithmic growth.

2.3 Exact order derivatives and a Stieltjes finite part

For later comparison with antiderivatives, set

$$\Gamma(1-w) = \sum_{j=0}^{\infty} g_j w^j, \quad g_0 = 1, \quad g_1 = \gamma, \quad g_2 = \frac{\gamma^2 + \zeta(2)}{2}. \quad (2.17)$$

For $L = \log(1/t)$ define the polynomial

$$P_n(L) = (-1)^{n+1} n! \sum_{j=0}^{n+1} g_j \frac{(-L)^{n+1-j}}{(n+1-j)!}. \quad (2.18)$$

Its leading term is $L^{n+1}/(n+1)$. The first three are

$$\begin{aligned} P_0(L) &= L - \gamma, \\ P_1(L) &= \frac{L^2}{2} - \gamma L + \frac{\gamma^2 + \zeta(2)}{2}, \\ P_2(L) &= \frac{L^3}{3} - \gamma L^2 + (\gamma^2 + \zeta(2))L - \frac{\gamma^3 + 3\gamma\zeta(2) + 2\zeta(3)}{3}. \end{aligned} \quad (2.19)$$

Proposition 2.6 (Convergent finite-part identity). *For $a > 0$, $n \geq 0$, and $0 < t < 2\pi$,*

$$e^{-at} \sum_{m=0}^{\infty} \frac{e^{-mt} \log^n(m+a)}{m+a} = P_n(\log(1/t)) + \gamma_n(a) + (-1)^n \sum_{j=1}^{\infty} \zeta^{[n]}(1-j, a) \frac{(-t)^j}{j!}. \quad (2.20)$$

The final series is convergent. The identity, including fixed a derivatives, holds locally uniformly. In particular the finite part at $t = 0$ is $\gamma_n(a)$, and the error after removing it is $O(t)$ on compact positive a intervals.

Proof. Put $k = 0$ in the paired form of (2.2), expand at $s = 1 + w$, and extract $(-1)^n n! [w^n]$. The coefficient of $[1 - e^{-Lw} \Gamma(1-w)]/w$ is exactly $P_n(L)$. The $j \geq 1$ terms contain no pole near $w = 0$, so Cauchy coefficient extraction commutes with their normally convergent series. $\square$

The proposition is an explicit generalized-shift version of the classical integer-order derivative calculus, not a replacement for the new uniform theorem. It supplies a useful numerical and conceptual bridge: the subtraction is independent of a , so differentiation in a removes it. The unrenormalized $n = 0$ Lerch value diverges at its endpoint, whereas its finite part and parameter derivatives have canonical Hurwitz limits.

2.4 Antiderivatives and exact logarithmic sum differences

For $a, b > 0$, subtracting (2.20) at the two shifts cancels P_n . In particular,

$$\gamma_n(a) - \gamma_n(b) = \lim_{t \downarrow 0} \sum_{m=0}^{\infty} \left[\frac{e^{-(m+a)t} \log^n(m+a)}{m+a} - \frac{e^{-(m+b)t} \log^n(m+b)}{m+b} \right]. \quad (2.21)$$

For fixed a, b , the unweighted difference series also converges absolutely, by the mean-value theorem and the bound $O((1 + \log^n m)/m^2)$ for its summand. The differently shifted exponential weights have the same limit: differentiate their summand in the shift variable and use $te^{-(m+v)t} \leq 1/[e(m+v)]$. This supplies the same summable bound uniformly in $t > 0$ on the interval between a and b , so dominated convergence applies. The integral in the shift variable is

$$\int_b^a \gamma_n(v) dv = \frac{(-1)^{n+1}}{n+1} [\zeta^{[n+1]}(0, a) - \zeta^{[n+1]}(0, b)]. \quad (2.22)$$

To see this without assigning values at a pole, integrate $G_v(w)$ first:

$$\int_b^a G_v(w) dv = \frac{\zeta(w, b) - \zeta(w, a) - (a - b)}{w}. \quad (2.23)$$

The numerator vanishes at zero because $\zeta(0, v) = 1/2 - v$; coefficient comparison gives (2.22). At $n = 0$ this is $-\log \Gamma(a) + \log \Gamma(b)$. At $n = 1$ it is half the difference of the second spectral derivatives at zero. These known primitive identities are part of the manuscript's Hurwitz calculus [3]; the new uniform resonance expansion supplies compatible limits when integer order and argument are both changing.

Remark 2.7 (A probability interpretation). For $k = 0$ and fixed a , the positive weights $e^{-(m+a)t}/(m+a)$, normalized by their sum, define a probability law on $\log(m+a)/\log(1/t)$. Its Laplace transforms tend to $h(u)$ for real u in a neighborhood of zero, so the law converges to the uniform distribution on $[0, 1]$. This follows directly from Theorem 2.3 and the standard uniqueness theorem for moment generating functions. The inverse logarithmic corrections explain exactly where the Stieltjes constants and gamma derivatives enter this limit. This interpretation is not needed in any proof below.

3 Periodic Hurwitz correlations and the gamma ladder

The polynomial moments in the repository describe one way to pair a Hurwitz jet with a test function. A second useful pairing is translation on the circle. It keeps the complete shift variable and produces polylogarithms, rather than only their values at unity. Its most useful consequence below is an exact convergent expansion proving strict convexity of the periodic autocorrelation of $\log \Gamma$.

The underlying Fourier calculation is classical. In particular, the zero-shift product integral is Theorem 3.1 of Espinosa–Moll [10]; their Example 7.2 gives the familiar square moment of $\log \Gamma$. The Fourier transform of Hurwitz zeta, Lerch's identity, and the local polylogarithm expansion are classical formulas [9]. The circle interpretation also belongs to the spectral framework discussed by Gay–Sebban [12]. We present the shifted calculus with its precise domains, derive a uniform family of Stieltjes-primitive cusp polynomials, and prove the convexity result. These are extensions developed here, without a claim of exhaustive bibliographic priority.

For $0 < x < 1$, let $\mathcal{Z}_s(x) = \zeta(s, x)$ and extend this function periodically, with its values at integers immaterial in integrals. Write $\{x\} = x - \lfloor x \rfloor$. For $0 < a < 1$ set

$$C(s, t; a) = \int_0^1 \zeta(s, x) \zeta(t, \{x + a\}) dx. \quad (3.1)$$

The two singularities occur at distinct points, $x = 0$ and $x = 1 - a$.

Theorem 3.1 (Shifted Hurwitz product). *For $0 < a < 1$, the integral (3.1) converges absolutely and is jointly holomorphic on $\operatorname{Re} s < 1$, $\operatorname{Re} t < 1$. On this domain,*

$$C(s, t; a) = \frac{\Gamma(1-s)\Gamma(1-t)}{(2\pi)^{2-s-t}} \left[e^{\pi i(t-s)/2} \operatorname{Li}_{2-s-t}(e^{2\pi i a}) + e^{\pi i(s-t)/2} \operatorname{Li}_{2-s-t}(e^{-2\pi i a}) \right]. \quad (3.2)$$

At $a = 0$, the same formula with both polylogarithms replaced by $\zeta(2-s-t)$ holds on the absolute-convergence domain

$$\operatorname{Re} s < 1, \quad \operatorname{Re} t < 1, \quad \operatorname{Re}(s+t) < 1. \quad (3.3)$$

All spectral derivatives can be taken under the integral on compact subsets of these domains.

Proof. The shift identity gives $\zeta(s, x) = x^{-s} + \zeta(s, 1+x)$. On every compact subset of $\operatorname{Re} s < 1$, this and its spectral derivatives are bounded near zero by a constant times $x^{-\sigma}(1 + |\log x|^r) + 1$, with some $\sigma < 1$. For a nonzero shift, the two endpoint singularities are separated. These bounds give absolute convergence, local domination, and joint holomorphy. At zero shift the extra condition in (3.3) controls the product x^{-s-t} .

For $\operatorname{Re} s < 0$ the classical Hurwitz Fourier series is absolutely convergent:

$$\mathcal{Z}_s(x) = \Gamma(1-s) \sum_{n \neq 0} (2\pi i n)^{s-1} e^{2\pi i n x}, \quad \log(2\pi i n) = \log(2\pi |n|) + \frac{\pi i}{2} \operatorname{sgn} n. \quad (3.4)$$

Multiplying the two series and integrating leaves the terms with opposite Fourier indices. Splitting the result into positive and negative indices proves (3.2) when $\operatorname{Re} s, \operatorname{Re} t < 0$. The right side is holomorphic on the larger product of half-planes: the order dependence of $\operatorname{Li}_w(z)$ is entire for fixed $z = e^{\pm 2\pi i a} \neq 1$, and the gamma factors have no poles there. Holomorphic continuation proves the assertion. The zero-shift proof uses the same argument on the connected domain (3.3). $\square$

The domain distinction matters: a nonzero shift permits $\operatorname{Re} s$ and $\operatorname{Re} t$ arbitrarily close to 1 individually, whereas coincident singularities impose an additional sum condition.

3.1 A factorization through one Hurwitz order

Put $u = s + t$ and define the holomorphic germ

$$F(s, t) = \frac{\Gamma(1-s)\Gamma(1-t)}{\Gamma(2-s-t)} \frac{\cos(\pi(s-t)/2)}{\cos(\pi(s+t)/2)}. \quad (3.5)$$

For example, F is holomorphic when $|s| + |t| < 1$, and $F(0, 0) = 1$. Let $C^+(s, t; a) = \frac{1}{2}[C(s, t; a) + C(t, s; a)]$.

Proposition 3.2 (Symmetric factorization). *In a neighborhood of $(s, t) = (0, 0)$, for $0 < a < 1$,*

$$C^+(s, t; a) = -\frac{F(s, t)}{2} [\zeta(u-1, a) + \zeta(u-1, 1-a)], \quad (3.6)$$

$$C(s, t; 0) = -F(s, t)\zeta(u-1). \quad (3.7)$$

Proof. Symmetrizing (3.2) produces

$$\frac{\Gamma(1-s)\Gamma(1-t)}{(2\pi)^{2-u}} \cos \frac{\pi(s-t)}{2} [\text{Li}_{2-u}(e^{2\pi ia}) + \text{Li}_{2-u}(e^{-2\pi ia})].$$

Apply the even part of (3.4) at order $u-1$:

$$\zeta(u-1, a) + \zeta(u-1, 1-a) = -\frac{2\Gamma(2-u) \cos(\pi u/2)}{(2\pi)^{2-u}} [\text{Li}_{2-u}(e^{2\pi ia}) + \text{Li}_{2-u}(e^{-2\pi ia})].$$

This proves the first formula. Letting $a \downarrow 0$ proves the second. The zero-shift formula is also exactly the beta-function form of the classical Espinosa–Moll product integral. $\square$

3.2 All spectral jets and Stieltjes antiderivatives

For $r \geq 0$, define

$$J_r(x) = \partial_s^r \zeta(s, x)|_{s=0}. \quad (3.8)$$

Then $J_0(x) = \frac{1}{2} - x$ and $J_1(x) = \log \Gamma(x) - \frac{1}{2} \log(2\pi)$. All J_r belong to $L^p(0, 1)$ for every finite p , since

$$J_r(x) = (-\log x)^r + \zeta^{(r)}(0) + O(x) \quad (x \downarrow 0). \quad (3.9)$$

Their means vanish: differentiate $\int_0^1 \zeta(s, x) dx = 0$ near $s = 0$.

The Stieltjes convention is $\zeta(1+v, x) = v^{-1} + \sum_{n \geq 0} (-1)^n \gamma_n(x) v^n / n!$. The classical argument-derivative identity gives

$$J'_{n+1}(x) = (-1)^{n+1} (n+1) \gamma_n(x), \quad \mathcal{G}_n(x) := \int_1^x \gamma_n(y) dy = \frac{(-1)^{n+1}}{n+1} [J_{n+1}(x) - \zeta^{(n+1)}(0)]. \quad (3.10)$$

Thus all correlations of these antiderivatives are mixed spectral derivatives of (3.2) at $s = t = 0$. This identifies their polylogarithm order derivatives without numerical integer-relation searches.

Theorem 3.3 (Universal cusp polynomials and a convergent expansion). *For nonnegative integers r, m , let*

$$E_{r,m}(a) = \int_0^1 [J_r(\{x+a\}) - J_r(x)][J_m(\{x+a\}) - J_m(x)] dx. \quad (3.11)$$

For $0 < a < 1$, with $L_a = \log(1/a)$,

$$E_{r,m}(a) = a P_{r,m}(L_a) + 2 \sum_{j \geq 1} \frac{a^{2j}}{(2j)!} \partial_s^r \partial_t^m [F(s, t)(u-1)_{2j} \zeta(u+2j-1)]|_{s=t=0}, \quad (3.12)$$

where $(z)_n = z(z+1) \cdots (z+n-1)$ and

$$P_{r,m}(L) = \partial_s^r \partial_t^m [F(s, t) e^{L(s+t)}]|_{s=t=0}. \quad (3.13)$$

The series converges locally uniformly for $0 < a < 1$. Each $P_{r,m}$ is a monic polynomial of degree $r+m$, and $E_{r,m}(a) = a P_{r,m}(L_a) + O(a^2)$ as $a \downarrow 0$. The apparent pole in the $j=1$ coefficient is removable.

Proof. Translation invariance and exchange of the two indices give

$$E_{r,m}(a) = 2 \partial_s^r \partial_t^m [C(s, t; 0) - C^+(s, t; a)] \Big|_{s=t=0}.$$

Use (3.6)–(3.7), then the shift identity and the Taylor series at Hurwitz parameter 1:

$$\begin{aligned} & \zeta(u-1, a) + \zeta(u-1, 1-a) - 2\zeta(u-1) \\ &= a^{1-u} + 2 \sum_{j \geq 1} \frac{(u-1)_{2j}}{(2j)!} \zeta(u+2j-1) a^{2j}. \end{aligned}$$

The Taylor radius is 1, the distance from parameter 1 to its nearest singularity 0. This proves convergence and permits spectral differentiation on smaller compact sets. At $j = 1$, $(u-1)_2 = u(u-1)$ cancels the pole of $\zeta(1+u)$. The singular term is $aF(s, t)e^{La(s+t)}$, giving (3.13). Its leading coefficient is $F(0, 0) = 1$. The remaining series starts at a^2 . $\square$

All coefficients in the cusp polynomials are explicit. A convenient closed generating formula is

$$\begin{aligned} \log F(s, t) = & -\log(1-s-t) + \sum_{n \geq 2} \frac{\zeta(n)}{n} [s^n + t^n - (s+t)^n] \\ & + \sum_{\substack{n \geq 2 \\ n \text{ even}}} \frac{2(1-2^{-n})\zeta(n)}{n} [(s+t)^n - (s-t)^n]. \end{aligned} \quad (3.14)$$

It follows from the gamma and cosine products and converges near zero. For instance,

$$P_{1,1}(L) = L^2 + 2L + 2 + \frac{\pi^2}{3}, \quad (3.15)$$

$$P_{1,2}(L) = L^3 + 3L^2 + \left(6 + \frac{2\pi^2}{3}\right)L + 6 + \frac{2\pi^2}{3} - 2\zeta(3), \quad (3.16)$$

$$\begin{aligned} P_{2,2}(L) = & L^4 + 4L^3 + \left(12 + \frac{4\pi^2}{3}\right)L^2 + \left(24 + \frac{8\pi^2}{3} - 8\zeta(3)\right)L \\ & + 24 + \frac{8\pi^2}{3} - 8\zeta(3) + \frac{7\pi^4}{45}. \end{aligned} \quad (3.17)$$

In particular the first nontrivial Stieltjes primitive satisfies

$$\int_0^1 [\mathcal{G}_1(\{x+a\}) - \mathcal{G}_1(x)]^2 dx = \frac{a}{4} P_{2,2}(\log(1/a)) + O(a^2).$$

The exact higher coefficients in (3.12) involve Stieltjes constants at the $j = 1$ resonance and zeta jets at positive odd integers for $j \geq 2$.

4 Strict convexity of the periodic log-gamma correlation

Write

$$\begin{aligned} g(x) &= \log \Gamma(x), \quad \ell = \log(2\pi), \quad A = \gamma + \ell, \\ R(a) &= \int_0^1 g(x)g(\{x+a\}) dx, \quad S(a) = \int_0^1 [g(\{x+a\}) - g(x)]^2 dx. \end{aligned}$$

Both extend continuously to the circle, and $S(a) = 2[R(0) - R(a)]$ and $S(1-a) = S(a)$. Here the square in S is an ordinary square of real-valued functions.

Proposition 4.1 (Two exact evaluations of the correlation). *For $0 < a < 1$, set $B_2(a) = a^2 - a + 1/6$ and write $\text{Li}_w^{[j]}(z) = \partial_w^j \text{Li}_w(z)$ for spectral derivatives. Then*

$$R(a) = \frac{\ell^2}{4} + \frac{1}{2\pi^2} \text{Re} \left[\text{Li}_2^{[2]}(e^{2\pi ia}) - 2A \text{Li}_2^{[1]}(e^{2\pi ia}) + \left(A^2 + \frac{\pi^2}{4} \right) \text{Li}_2(e^{2\pi ia}) \right] \quad (4.1)$$

$$= \frac{\ell^2}{4} - \frac{\zeta''(-1, a) + \zeta''(-1, 1-a)}{2} - \zeta'(-1, a) - \zeta'(-1, 1-a) + \left(1 + \frac{\pi^2}{6} \right) B_2(a). \quad (4.2)$$

Proof. Since $g = J_1 + \ell/2$ and $\int J_1 = 0$, the desired centered correlation is $\partial_s \partial_t C(s, t; a)|_{0,0}$. Apply these derivatives to (3.2). The mixed derivative of the phase contributes $\pi^2/4$, giving (4.1). For (4.2), use the symmetric factorization, $F_s(0,0) = F_t(0,0) = 1$, $F_{st}(0,0) = 2 + \pi^2/3$, and $\zeta(-1, a) + \zeta(-1, 1-a) = -B_2(a)$. $\square$

Theorem 4.2 (Convergent odd-zeta expansion and strict concavity). *For every $0 < a < 1$,*

$$S(a) = a \left[\log^2 \frac{1}{a} + 2 \log \frac{1}{a} + 2 + \frac{\pi^2}{3} \right] + \left(2\gamma_1 - \frac{\pi^2}{3} \right) a^2 - \sum_{j=2}^{\infty} \frac{2a^{2j}}{j(2j-1)} [H_{2j-2} \zeta(2j-1) + \zeta'(2j-1)]. \quad (4.3)$$

This is a convergent identity, not only an asymptotic expansion. Every bracket in the series is strictly positive. The function S is strictly concave on $(0,1)$, so R is strictly convex there. Consequently S has a unique maximum, and R a unique minimum, at $a = 1/2$.

Proof. Specialize Theorem 3.3 to $r = m = 1$. For the quadratic term,

$$u(u-1)\zeta(1+u) = -1 + (1-\gamma)u + (\gamma + \gamma_1)u^2 + O(u^3).$$

Its product with $F(s, t)$ has mixed derivative $-(2 + \pi^2/3) + 2(1-\gamma) + 2(\gamma + \gamma_1) = 2\gamma_1 - \pi^2/3$. For $j \geq 2$, put $M = 2j - 2$. Factoring out the simple zero at $u = 0$ gives

$$(u-1)_{2j} = -M! u [1 + (H_M - 1)u + O(u^2)].$$

Since $F_s = F_t = 1$ at the origin, the mixed derivative of $F(s, t)(u-1)_{2j}\zeta(u+2j-1)$ is

$$-2M! [H_M \zeta(2j-1) + \zeta'(2j-1)].$$

Insertion into (3.12) yields (4.3).

We give analytic sign bounds, so the qualitative conclusion does not depend on numerical values of Stieltjes or zeta constants. For $v \geq 3$, the absolutely convergent series gives

$$0 < -\zeta'(v) = \sum_{n \geq 2} \frac{\log n}{n^v} < \sum_{n \geq 2} \frac{n-1}{n^v} \leq \sum_{n \geq 2} \frac{1}{n^2} < 1.$$

The last inequality follows by comparison with $\int_1^\infty x^{-2} dx = 1$. But $H_{2j-2} \zeta(2j-1) > H_2 = 3/2$ for $j \geq 2$. Thus every series bracket is positive.

For the quadratic coefficient, let $f(x) = \log x/x$. The defining limit for the first Stieltjes constant is

$$\gamma_1 = \lim_{N \rightarrow \infty} \left[\sum_{n=1}^N f(n) - \int_1^N f(x) dx \right].$$

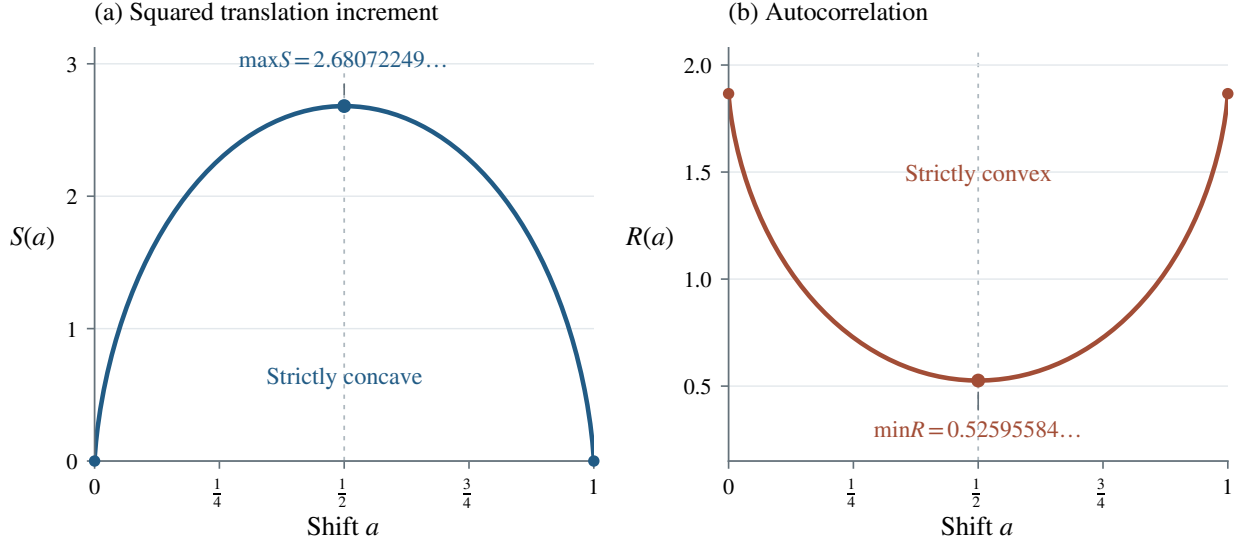


Figure 1: Periodic log-Gamma correlations. The squared translation increment $S(a)$ is strictly concave, and the autocorrelation $R(a) = R(0) - S(a)/2$ is strictly convex. The dashed guides mark the unique interior extrema at $a = 1/2$. The curves use the convergent series of Theorem 4.2 and analytic endpoint limits; the shape assertions are proved above.

The absolute value of the expression in brackets is at most $f(N) + \int_1^N |f'(x)| dx$. As $N \rightarrow \infty$, this proves $|\gamma_1| \leq 2/e$. Hence $2\gamma_1 - \pi^2/3 \leq 4/e - \pi^2/3 < 0$.

The second derivative of the leading term of (4.3) is $-2\log(1/a)/a < 0$ on $(0, 1)$. The quadratic term has negative second derivative, and every remaining term has negative second derivative. Local uniform convergence permits differentiating the series twice. Therefore $S''(a) < 0$. Symmetry $S(1-a) = S(a)$ and strict concavity give the unique maximum at $1/2$. The statement about R follows from $S = 2(R(0) - R)$. $\square$

Corollary 4.3 (Sharp endpoint expansion). *As $a \downarrow 0$,*

$$S(a) = a \left[\log^2 \frac{1}{a} + 2 \log \frac{1}{a} + 2 + \frac{\pi^2}{3} \right] + \left(2\gamma_1 - \frac{\pi^2}{3} \right) a^2 - \frac{a^4}{3} \left[\frac{3}{2} \zeta(3) + \zeta'(3) \right] + O(a^6). \quad (4.4)$$

The leading behavior is $S(a) \sim a \log^2(1/a)$, with the full coefficient polynomial and the first analytic corrections explicit.

Corollary 4.4 (Effective truncation). *Let $S_N(a)$ be the right side of (4.3) truncated after $j = N$, where $N \geq 2$. Then*

$$0 < S_N(a) - S(a) \leq \frac{2\zeta(3)[1 + \log(2N + 2)]}{(N + 1)^2} \frac{a^{2N+2}}{1 - a^2}. \quad (4.5)$$

By symmetry one can always replace a by $\min(a, 1-a) \leq 1/2$ before evaluating this series.

Proof. The omitted coefficients have one sign. For $j \geq 2$,

$$0 < H_{2j-2}\zeta(2j-1) + \zeta'(2j-1) < \zeta(3)[1 + \log(2j)], \quad j(2j-1) \geq j^2.$$

The function $(1 + \log(2x))/x^2$ decreases for $x \geq 1$. Majorizing it at $x = N + 1$ and summing the geometric tail proves (4.5). $\square$

4.1 The extremal half-shift and a geometric zeta identity

Corollary 4.5 (Exact maximum). *Let $q = \log 2$ and $A = \gamma + \log(2\pi)$. Then*

$$\begin{aligned} S(1/2) &= \frac{3\zeta''(2) + 2(q - 3A)\zeta'(2)}{2\pi^2} + \frac{A^2}{4} - \frac{Aq}{6} - \frac{q^2}{12} + \frac{\pi^2}{16} \\ &= 2.68072248798050738482152413952\dots \end{aligned} \quad (4.6)$$

Equating this value with the convergent shift expansion gives

$$\sum_{j=2}^{\infty} \frac{2[H_{2j-2}\zeta(2j-1) + \zeta'(2j-1)]}{4^j j(2j-1)} = 1 + \log 2 + \frac{\log^2 2}{2} + \frac{\pi^2}{12} + \frac{\gamma_1}{2} - S(1/2). \quad (4.7)$$

Proof. In (4.1), use $\text{Li}_w(-1) = (2^{1-w} - 1)\zeta(w)$. Hence

$$S(1/2) = \frac{1}{\pi^2} \left[(\partial_w - A)^2 + \frac{\pi^2}{4} \right] [(2 - 2^{1-w})\zeta(w)] \Big|_{w=2}.$$

At $w = 2$, the three derivatives of $2 - 2^{1-w}$ required here are $3/2$, $q/2$, and $-q^2/2$. Expanding and using $\zeta(2) = \pi^2/6$ proves (4.6). Substitution of $a = 1/2$ into (4.3) proves (4.7). $\square$

5 Balanced negative polygamma correlations

For an integer $m \geq 1$, define the classical balanced normalization

$$\Psi_m(x) = \frac{\zeta'(1-m, x) + H_{m-1}\zeta(1-m, x)}{(m-1)!}. \quad (5.1)$$

This is the Espinosa–Moll balanced negapolygamma [11]. In particular $\Psi_1 = J_1 = g - \ell/2$. It differs from the zero-based repeated integrals of g used elsewhere in the manuscript by explicit polynomials. The normalization must be specified before periodizing: polynomial changes can change endpoint jumps and Fourier decay.

For $m \geq 2$, $\Psi'_m = \Psi_{m-1}$, $\int_0^1 \Psi_m = 0$, and the endpoint values match. The same mean-zero assertion holds for $m = 1$, whose left endpoint is logarithmically singular. These facts follow directly from the Hurwitz shift and derivative identities.

Theorem 5.1 (One polylogarithmic operator for the integrated tower). *For all integers $m, n \geq 1$ and $0 < a < 1$,*

$$\begin{aligned} &\int_0^1 \Psi_m(x) \Psi_n(\{x+a\}) dx \\ &= \frac{2}{(2\pi)^{m+n}} \text{Re} \left[e^{\pi i(m-n)/2} \left((\partial_w - A)^2 + \frac{\pi^2}{4} \right) \text{Li}_w(e^{2\pi i a}) \right]_{w=m+n}, \quad A = \gamma + \log(2\pi). \end{aligned} \quad (5.2)$$

The zero-shift value is obtained by replacing Li_w with $\zeta(w)$. At every rational shift, all terms reduce by a finite Fourier transform to Hurwitz zeta values and their first two spectral derivatives at the positive integer $m+n$.

Proof. Differentiate (3.4) at $s = 1 - m$ and add the H_{m-1} multiple. Since $\psi(m) = H_{m-1} - \gamma$, the harmonic numbers cancel, giving

$$\widehat{\Psi}_m(k) = \frac{\gamma + \log(2\pi i k)}{(2\pi i k)^m}, \quad k \neq 0; \quad \widehat{\Psi}_m(0) = 0. \quad (5.3)$$

For $m = 1$ this is interpreted in L^2 ; for $m \geq 2$ the series is absolutely convergent. Parseval's product formula therefore applies in every case. For $k > 0$, the product of the opposite-index coefficients has modulus factor $(A + \log k)^2 + \pi^2/4$ and phase $e^{\pi i(n-m)/2}$. Combining both signs of k gives (5.2) because $\partial_w k^{-w} = -(\log k)k^{-w}$. The rational-shift statement follows by twice differentiating

$$\mathrm{Li}_w(e^{2\pi i p/q}) = q^{-w} \sum_{r=1}^q e^{2\pi i p r/q} \zeta(w, r/q).$$

□

Corollary 5.2 (Exact regularity threshold). *The periodized function Ψ_m belongs to the circle Sobolev space H^σ exactly when $\sigma < m - 1/2$. More generally, for any $k, r \geq 0$, the periodized function $x \mapsto \partial_s^r \zeta(-k, x)$ belongs to H^σ exactly when $\sigma < k + 1/2$.*

Proof. With the Fourier definition $\|f\|_{H^\sigma}^2 \asymp \sum_k (1 + k^2)^\sigma |\widehat{f}(k)|^2$, (5.3) reduces the first claim to convergence of $\sum_{k \geq 2} k^{2\sigma-2m} \log^2 k$. For a general spectral jet, differentiating (3.4) gives

$$|\partial_s^r \widehat{\zeta(-k, \cdot)}(n)| \sim \frac{k!}{(2\pi)^{k+1}} \frac{(\log n)^r}{n^{k+1}} \quad (n \rightarrow +\infty),$$

with the same power law at negative frequencies. The comparison series converges exactly below the stated threshold, including divergence at equality. □

Remark 5.3 (Boundary of ordinary integral formulas). The restriction $m, n \geq 1$ in Theorem 5.1 is essential. The next derivative is the ordinary digamma function, with a $1/x$ singularity; periodic shifted products containing it need an explicitly prescribed finite-part or distributional interpretation. The absolutely convergent integral theorem does not silently provide such a regularization.

6 A uniform transition for high Herglotz derivatives

The frozen manuscript proves a sharp asymptotic expansion for the Herglotz derivatives when their order tends to infinity at a fixed positive argument [5]. Its final research paragraph explicitly leaves open a joint limit in which the argument also varies. We resolve the transition scale $x \asymp r$. The result has a complete expansion with computable coefficients, an explicit finite-order remainder, and a profile that admits arithmetic, Bessel, Gamma, and dilogarithmic representations. The special-function identities used in the proof are classical; the contribution here is the joint-limit theorem and its systematic connection to this profile. No historical priority is claimed for an isolated transformed divisor sum.

Let

$$F(x) = \sum_{n \geq 1} \frac{\psi(nx) - \log(nx)}{n}, \quad D_r(x) = \frac{(-1)^{r+1} x^{r+1}}{r!} F^{(r)}(x), \quad x > 0, \quad r \geq 1. \quad (6.1)$$

Here $F^{(r)}$ differentiates the argument x . Define

$$B(t) = \frac{1}{1 - e^{-t}} - \frac{1}{t}, \quad B(0) = \frac{1}{2}, \quad S(\lambda) = \sum_{n \geq 1} \frac{1}{n^2} B\left(\frac{1}{n\lambda}\right), \quad \lambda > 0. \quad (6.2)$$

The letter B without an index denotes this elementary function; B_j with an integer index denotes a Bernoulli number.

6.1 An exact probability representation and a finite remainder

Lemma 6.1 (Positive kernel and derivative bounds). *For $t > 0$, $1/2 < B(t) < 1$ and $B'(t) > 0$. For every integer $q \geq 1$,*

$$\sup_{t \geq 0} |B^{(q)}(t)| \leq \frac{2q! \zeta(q+1)}{(2\pi)^{q+1}}. \quad (6.3)$$

Proof. The hyperbolic partial-fraction expansion [14, Eq. 4.36.3] gives

$$B(t) = \frac{1}{2} + 2t \sum_{m \geq 1} \frac{1}{t^2 + 4\pi^2 m^2} = \frac{1}{2} + \sum_{m \geq 1} \left(\frac{1}{t - 2\pi i m} + \frac{1}{t + 2\pi i m} \right). \quad (6.4)$$

The second sum is paired as displayed. For $q \geq 1$, differentiating gives an absolutely and uniformly convergent series on the real axis; $|t \pm 2\pi i m| \geq 2\pi m$ proves (6.3). Furthermore,

$$B'(t) = \frac{1}{t^2} - \frac{1}{4 \sinh^2(t/2)} > 0,$$

since $\sinh(t/2) > t/2$. The limits at zero and infinity are $1/2$ and 1 , respectively. The derivative bound is asserted only for $q \geq 1$; its displayed right-hand side would be divergent at $q = 0$. $\square$

Let Y_r have the gamma density

$$\frac{r^{r+1}}{r!} y^r e^{-ry}, \quad y > 0, \quad (6.5)$$

and put

$$\mu_m(r) = \mathbb{E}(Y_r - 1)^m = \sum_{\ell=0}^m (-1)^{m-\ell} \binom{m}{\ell} \frac{(r+1)^\ell}{r^\ell}. \quad (6.6)$$

Each $\mu_m(r)$ is an exact polynomial in r^{-1} ; it need not be evaluated by subtracting floating-point numbers.

Theorem 6.2 (Exact mixture and computable Taylor certificate). *For all $r \geq 1$ and $\lambda > 0$,*

$$D_r(r\lambda) = \mathbb{E}S(\lambda/Y_r). \quad (6.7)$$

Write $h_\lambda(y) = S(\lambda/y)$ for $y > 0$, extended by $h_\lambda(0) = \zeta(2)/2$. For $J \geq 0$ and $q = 2J + 2$,

$$\begin{aligned} \left| D_r(r\lambda) - \sum_{m=0}^{2J+1} \frac{h_\lambda^{(m)}(1)}{m!} \mu_m(r) \right| \\ \leq \frac{2\zeta(q+1)\zeta(q+2)}{(2\pi)^{q+1}\lambda^q} \mu_q(r). \end{aligned} \quad (6.8)$$

This is a finite- r inequality, valid on the entire positive λ axis, rather than an unspecified asymptotic error.

Proof. The classical digamma integral, also used in the Herglotz theory of Radchenko–Zagier [13], is

$$\psi(z) - \log z = - \int_0^\infty e^{-zt} B(t) dt, \quad z > 0.$$

Since $0 < B < 1$, the defining sum and any fixed number of its argument derivatives converge locally uniformly. Differentiating and substituting $u = nxt$ gives

$$D_r(x) = \sum_{n \geq 1} \frac{1}{n^2 r!} \int_0^\infty u^r e^{-u} B\left(\frac{u}{nx}\right) du. \quad (6.9)$$

All integrands are positive and bounded after normalization by the summable sequence n^{-2} . Thus every interchange is justified, and $x = r\lambda$ proves (6.7).

From (6.3), for every $q \geq 1$,

$$\sup_{y \geq 0} |h_\lambda^{(q)}(y)| \leq \frac{2q! \zeta(q+1) \zeta(q+2)}{(2\pi)^{q+1} \lambda^q}.$$

Taylor's formula at $y = 1$, through degree $q - 1$, therefore has a global remainder bounded by the last constant times $|y - 1|^q / q!$. Taking expectations proves (6.8), because q is even. Formula (6.6) follows from the elementary gamma moments. $\square$

6.2 The complete expansion and its scope

Define the finite polynomials $C_j(z) = \sum_m c_{j,m} z^m$ by

$$\sum_{j \geq 0} C_j(z) w^j = \exp \left\{ \sum_{\ell \geq 1} w^\ell \left(\frac{z^\ell}{\ell} + \frac{z^{\ell+1}}{\ell+1} \right) \right\}. \quad (6.10)$$

Thus $\deg C_j \leq 2j$. Let $\mathcal{E} = \lambda d/d\lambda$ and use the rising operator factorial $\mathcal{E}^{\overline{m}} = \mathcal{E}(\mathcal{E}+1) \cdots (\mathcal{E}+m-1)$, with value 1 at $m = 0$. Put

$$\mathcal{T}_j = \sum_m c_{j,m} (-1)^m \mathcal{E}^{\overline{m}}. \quad (6.11)$$

This definition specifies every coefficient without an implicit recurrence for special-function values.

Theorem 6.3 (Uniform complete transition expansion). *For any nonnegative integers J, u and any compact interval $I \subset (0, \infty)$,*

$$D_r(r\lambda) = \sum_{j=0}^J \frac{(\mathcal{T}_j S)(\lambda)}{r^j} + O_{I,J,u}(r^{-J-1}), \quad r \rightarrow \infty, \quad (6.12)$$

uniformly for $\lambda \in I$, also after up to u derivatives in λ . In particular,

$$\begin{aligned} D_r(r\lambda) &= S(\lambda) + \frac{\lambda^2 S''(\lambda)}{2r} \\ &\quad + \frac{1}{r^2} \left(\frac{\lambda^2 S''(\lambda)}{2} + \frac{2\lambda^3 S'''(\lambda)}{3} + \frac{\lambda^4 S^{(4)}(\lambda)}{8} \right) + O_I(r^{-3}). \end{aligned} \quad (6.13)$$

Proof. The exact moment generating function is

$$\mathbb{E}e^{z(Y_r-1)} = e^{-z}(1 - z/r)^{-r-1}.$$

Its logarithm expands as the exponent in (6.10) with $w = r^{-1}$. Therefore the coefficient of r^{-j} in $\mu_m(r)/m!$ is $[z^m]C_j(z)$. In particular $\mu_{2J+2}(r) = O_J(r^{-J-1})$. The finite Taylor certificate proves the complete expansion, since all central moments in it are exact polynomials. Finally,

$$h_\lambda^{(m)}(1) = (-1)^m \mathcal{E}^{\overline{m}} S(\lambda),$$

which follows by induction on m , proves (6.11).

For derivative uniformity, apply the same argument to each fixed $\partial_\lambda^v h_\lambda$. For $b \geq 1$ the function $t^b B^{(b)}(t)$ is bounded on $[0, \infty)$, by regularity at zero and the elementary large- t form $B(t) = 1 - t^{-1} + O(e^{-t})$. Thus the required parameter derivatives may also be passed through the expectation and the n sum. A mixed derivative is a finite sum of terms involving $t^b B^{(q+b)}(t)$, with $q = 2J + 2 \geq 2$. Differentiating the paired fractions in (6.4) gives

$$\sup_{t \geq 0} |t^b B^{(q+b)}(t)| \leq \frac{2(q+b)! \zeta(q+1)}{(2\pi)^{q+1}}.$$

Indeed $t^b \leq |t \pm 2\pi i m|^b$. The remaining powers of n^{-1} are summable and λ^{-1} is bounded on I . This supplies the required uniform Taylor remainders after each parameter differentiation. No uncontrolled big- O term is differentiated.

For the displayed coefficients, $C_1(z) = z + z^2/2$ and $C_2(z) = z^2 + 5z^3/6 + z^4/8$; substituting into (6.11) gives (6.13). $\square$

Corollary 6.4 (All limiting ratios and noncommuting limits). *The function S is strictly decreasing, with*

$$\lim_{\lambda \downarrow 0} S(\lambda) = \zeta(2), \quad \lim_{\lambda \rightarrow \infty} S(\lambda) = \frac{\zeta(2)}{2}.$$

If $r \rightarrow \infty$ through integers and $x_r/r \rightarrow \lambda$ in the extended interval $[0, \infty]$, then $D_r(x_r)$ tends to the corresponding value of this continuously extended profile. Consequently,

$$\lim_{x \rightarrow \infty} \lim_{r \rightarrow \infty} D_r(x) = \zeta(2), \quad \lim_{r \rightarrow \infty} \lim_{x \rightarrow \infty} D_r(x) = \frac{\zeta(2)}{2}.$$

The complete correction expansion is claimed on compact positive ratio intervals; the two endpoint statements concern the leading limit only.

Proof. Strict monotonicity follows termwise from $B' > 0$. Dominated convergence, using $1/2 \leq B \leq 1$, proves the limits of S . Moreover $Y_r \rightarrow 1$ in probability, since $\mathbb{E}(Y_r - 1)^2 = r^{-1} + 2r^{-2}$. In (6.9) each fixed- n argument consequently tends in probability to $1/(n\lambda)$, to infinity, or to zero, according to the ratio. Bounded convergence in probability and the summable n^{-2} bound give the asserted three cases. At fixed r , letting $x \rightarrow \infty$ in (6.9) gives $\zeta(2)/2$ directly. $\square$

7 Arithmetic continuation and zeros of the centered profile

Write $\sigma_{-1}(N) = \sum_{d|N} d^{-1}$. This section gives exact formulas for the function that enters the joint-limit theorem, including the complex zeros of its centered reciprocal form $H(y) - \zeta(2)/2$. These are zeros of that centered function, not zeros of S , H , or the original generalized Stieltjes constants.

Theorem 7.1 (Arithmetic representation and meromorphic zeros). *For $\lambda > 0$,*

$$S(\lambda) = \frac{\zeta(2)}{2} + 2\lambda \sum_{N \geq 1} \frac{\sigma_{-1}(N)}{1 + 4\pi^2 N^2 \lambda^2}. \quad (7.1)$$

The reciprocal profile $H(y) = S(1/y)$ extends meromorphically to the complex plane by

$$H(y) = \frac{\zeta(2)}{2} + 2y \sum_{N \geq 1} \frac{\sigma_{-1}(N)}{y^2 + 4\pi^2 N^2}. \quad (7.2)$$

It has simple poles at $y = \pm 2\pi i N$, of residue $\sigma_{-1}(N)$. The zeros of $H(y) - \zeta(2)/2$ are precisely the simple zero $y = 0$ and the simple pairs $y = \pm ib_N$, where there is exactly one b_N in every interval

$$2\pi N < b_N < 2\pi(N + 1), \quad N \geq 1. \quad (7.3)$$

There are no other zeros. In particular, every nonzero zero is purely imaginary and strictly interlaces with the poles.

Proof. Insert (6.4) into (6.2). All terms are positive for positive arguments, so rearrangement is permitted. Grouping the pair (m, n) by $N = mn$ gives (7.1). Since $\sigma_{-1}(N) \leq 1 + \log N$, the series in (7.2) converges locally uniformly away from its displayed poles, and direct residue calculation gives their residues.

Put

$$T(z) = \sum_{N \geq 1} \frac{\sigma_{-1}(N)}{z + 4\pi^2 N^2}, \quad H(y) - \zeta(2)/2 = 2yT(y^2).$$

For nonreal z ,

$$\operatorname{Im} T(z) = -(\operatorname{Im} z) \sum_{N \geq 1} \frac{\sigma_{-1}(N)}{|z + 4\pi^2 N^2|^2} \neq 0.$$

For real $z > -4\pi^2$, $T(z) > 0$. On each interval $(-4\pi^2(N + 1)^2, -4\pi^2 N^2)$ the derivative $T'(z) = -\sum \sigma_{-1}(j)/(z + 4\pi^2 j^2)^2$ is strictly negative, while its endpoint limits are $+\infty$ and $-\infty$. There is therefore exactly one simple zero in each interval and none elsewhere. Taking square roots proves (7.3). Finally $H'(0) = 2T(0) = \zeta(3)/12 > 0$, so the zero at the origin is simple as well. $\square$

The same representation gives positive moment and quadrature structures. For example, for $v > 0$,

$$\frac{S(\sqrt{v}) - \zeta(2)/2}{2\sqrt{v}} = \sum_{N \geq 1} \frac{\sigma_{-1}(N)}{1 + 4\pi^2 N^2 v}$$

is completely monotone in v : each derivative has the required alternating sign, and the differentiated series converges locally uniformly. This is a direct property of the displayed positive discrete measure.

7.1 Two complementary exact representations

Theorem 7.2 (Convergent zeta series and dual Bessel series). *For $\lambda > 1/(2\pi)$,*

$$S(\lambda) = \frac{\zeta(2)}{2} + \sum_{m \geq 1} \frac{B_{2m} \zeta(2m + 1)}{(2m)! \lambda^{2m-1}}, \quad (7.4)$$

an absolutely convergent series. For every $\lambda > 0$, put $R(\lambda) = S(\lambda) - \zeta(2) - \lambda(\log \lambda - \gamma)$. Then

$$R(\lambda) = 4\lambda \sum_{N \geq 1} \sigma_{-1}(N) \operatorname{Re} \left[\sqrt{\frac{2\pi i N}{\lambda}} K_1 \left(2\sqrt{\frac{2\pi i N}{\lambda}} \right) \right], \quad (7.5)$$

with principal square roots. The series is absolutely convergent and locally uniformly convergent on the positive λ axis.

Proof. The Taylor expansion of $B(t)$ at zero has radius 2π . Substitute it in (6.2); absolute convergence when $1/\lambda < 2\pi$ gives (7.4).

For the second identity, Mellin inversion of $(1 + u^2)^{-1}$ in (7.1) gives, for $1 < c < 2$,

$$S(\lambda) - \frac{\zeta(2)}{2} = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\pi \zeta(s) \zeta(s+1)}{\sin(\pi s/2)} (2\pi)^{-s} \lambda^{1-s} ds. \quad (7.6)$$

On this line the Dirichlet series is absolutely convergent; the sine denominator supplies exponential vertical decay. Move the line to $\operatorname{Re} s = -A$, with any fixed $A > 1$. The only poles crossed are $s = 1$ and $s = 0$. The negative even sine poles are cancelled by the trivial zeros of $\zeta(s)$. Standard vertical-strip bounds for zeta and the exponential factor justify the horizontal limits. The residue at 1 is $\zeta(2)/2$. At zero use $\zeta(0) = -1/2$, $\zeta'(0) = -\frac{1}{2} \log(2\pi)$, and $\zeta(1+s) = s^{-1} + \gamma + O(s)$; the residue is $\lambda(\log \lambda - \gamma)$.

The zeta functional equation [17], applied twice, transforms the remaining integral into

$$R(\lambda) = \frac{2\lambda}{2\pi i} \int_{(A)} \Gamma(w) \Gamma(w+1) \zeta(w) \zeta(w+1) \cos(\pi w/2) \left(\frac{\lambda}{2\pi} \right)^w dw. \quad (7.7)$$

Here $\zeta(w) \zeta(w+1) = \sum \sigma_{-1}(N) N^{-w}$ is absolutely convergent, and the product of the two gamma factors and the cosine decays exponentially on the line. Termwise integration is consequently justified. The standard Bessel Mellin–Barnes formula [15, Eq. 10.32.13] gives

$$\frac{1}{2\pi i} \int_{(A)} \Gamma(w) \Gamma(w+1) z^{-w} dw = 2\sqrt{z} K_1(2\sqrt{z}), \quad |\arg z| < \pi.$$

Taking the average of its values at $z = 2\pi i N/\lambda$ and its conjugate produces the cosine factor and proves (7.5).

For an explicit domination, set $v_N = 2\sqrt{\pi N/\lambda}$. The real-line Bessel integral implies $|K_1(w)| \leq K_1(\operatorname{Re} w) \leq K_{3/2}(\operatorname{Re} w)$ for $\operatorname{Re} w > 0$. The last inequality follows from $\cosh t \leq \cosh(3t/2)$ in that integral. Hence the absolute value of the N th summand in (7.5) is at most

$$2^{3/2} \pi^{3/4} \lambda^{3/4} \sigma_{-1}(N) N^{1/4} e^{-v_N} (1 + v_N^{-1}). \quad (7.8)$$

This proves absolute and local uniform convergence independently of the contour interchange. $\square$

Remark 7.3 (A simple numerical tail certificate). Put $a = 2\sqrt{\pi/\lambda}$ and $C = 2^{3/2} \pi^{3/4} \lambda^{3/4}$. For an integer cutoff $N \geq 1$ with $a\sqrt{N} \geq 3/2$, the absolute tail in (7.5) is bounded by

$$4C \left(1 + \frac{1}{a\sqrt{N}} \right) a^{-7/2} \Gamma \left(\frac{7}{2}, a\sqrt{N} \right), \quad (7.9)$$

where the final Gamma is the upper incomplete gamma function. Indeed $\sigma_{-1}(n) \leq 1 + \log n \leq 2\sqrt{n}$ for $n \geq 1$, and $t^{3/4} e^{-a\sqrt{t}}$ is decreasing for $a\sqrt{t} \geq 3/2$. Use the integral test and substitute $u = a\sqrt{t}$.

Corollary 7.4 (An exponentially small profile correction). *As $\lambda \downarrow 0$, let*

$$A(\lambda) = 2^{5/4} \pi^{3/4} \lambda^{3/4} e^{-2\sqrt{\pi/\lambda}}, \quad \phi(\lambda) = 2\sqrt{\pi/\lambda} - \frac{\pi}{8}.$$

Then

$$R(\lambda) = A(\lambda) \left[\cos \phi(\lambda) + \frac{3}{16} \sqrt{\frac{\lambda}{2\pi}} \cos \left(\phi(\lambda) + \frac{\pi}{4} \right) + O(\lambda) \right]. \quad (7.10)$$

This is additive; no division by a cosine near a zero is intended. There are complete further powers of $\sqrt{\lambda}$, supplied by the classical K_1 expansion. In particular R takes both signs arbitrarily close to zero.

Proof. In the $N = 1$ sector, use the standard large-argument expansion [15, Eq. 10.40.2]

$$K_1(2\sqrt{z}) = \frac{\sqrt{\pi}}{2z^{1/4}} e^{-2\sqrt{z}} \left[1 + \frac{3}{16\sqrt{z}} + O(z^{-1}) \right]$$

on the ray $\arg z = \pi/2$. Taking its real part gives exactly (7.10). From (7.8), the aggregate of the sectors $N \geq 2$ is $O(\lambda^{3/4} e^{-2\sqrt{2\pi/\lambda}})$: factor out its $N = 2$ exponential and dominate the remaining series at any fixed positive lower value of $2\sqrt{\pi/\lambda}$. This is smaller than every algebraic correction in the first sector. The phase passes through arbitrarily large multiples of π as $\lambda \downarrow 0$, proving the sign assertion. $\square$

8 Gamma and dilogarithm antiderivatives of the profile

The profile admits convergent primitive identities in which the subtractions are essential. They also provide useful test cases for symbolic integration across the Gamma and polylogarithmic descriptions.

Theorem 8.1 (Two normalized primitives). *For $y \geq 0$ define*

$$G(y) = - \sum_{n \geq 1} \frac{1}{n} \log \left[\Gamma \left(1 + \frac{iy}{2\pi n} \right) \Gamma \left(1 - \frac{iy}{2\pi n} \right) \right], \quad (8.1)$$

$$P(y) = \sum_{n \geq 1} \left\{ \text{Li}_2(e^{-y/n}) - \zeta(2) - \frac{y}{n} \log \frac{y}{n} + \frac{y}{n} + \frac{y^2}{4n^2} \right\}, \quad (8.2)$$

where terms at $y = 0$ are taken by continuity. Each Gamma product is strictly positive, so its logarithm is the real logarithm. Both series converge locally uniformly, with the derivatives needed below, and

$$P'(y) = G(y), \quad G'(y) = P''(y) = H(y) - \frac{\zeta(2)}{2}, \quad P(0) = P'(0) = G(0) = 0. \quad (8.3)$$

Equivalently,

$$\int_{\lambda}^{\infty} \frac{S(u) - \zeta(2)/2}{u^2} du = G(1/\lambda), \quad \lambda > 0. \quad (8.4)$$

For $|y| < 2\pi$ the corresponding convergent zeta-product series are

$$G(y) = \sum_{m \geq 1} \frac{B_{2m} \zeta(2m+1)}{(2m)!(2m)} y^{2m}, \quad (8.5)$$

$$P(y) = \sum_{m \geq 1} \frac{B_{2m} \zeta(2m+1)}{(2m)!(2m)(2m+1)} y^{2m+1}. \quad (8.6)$$

For example,

$$G(y) = \frac{\zeta(3)}{24}y^2 - \frac{\zeta(5)}{2880}y^4 + \cdots, \quad P(y) = \frac{\zeta(3)}{72}y^3 - \frac{\zeta(5)}{14400}y^5 + \cdots.$$

Proof. The reflection formula and the gamma recurrence [16, §5.5] give, for $t \geq 0$,

$$\Gamma\left(1 + \frac{it}{2\pi}\right) \Gamma\left(1 - \frac{it}{2\pi}\right) = \frac{t/2}{\sinh(t/2)}.$$

Consequently the function

$$L(t) = \log \frac{\sinh(t/2)}{t/2} = \frac{t}{2} + \log(1 - e^{-t}) - \log t$$

satisfies $L(0) = 0$ and $L'(t) = B(t) - 1/2$. Its primitive vanishing at zero is

$$J(t) = \text{Li}_2(e^{-t}) - \zeta(2) - t \log t + t + \frac{t^2}{4}, \quad J'(t) = L(t).$$

Indeed $d \text{Li}_2(e^{-t})/dt = \log(1 - e^{-t})$. Now $G(y) = \sum n^{-1}L(y/n)$ and $P(y) = \sum J(y/n)$. Since $L(t) = O(t^2)$, $J(t) = O(t^3)$, and their requisite derivatives have the corresponding bounds at zero, the series and their stated derivatives converge uniformly on bounded positive y intervals. This proves all of (8.3). The substitution $y = 1/u$ proves (8.4). Termwise integration of the Taylor series for $H(y) - \zeta(2)/2$ proves (8.5)–(8.6). $\square$

Further questions arising from this theorem. The complete expansion above suggests a uniform matching analysis when $\lambda = \lambda_r \downarrow 0$: one must compare its derivative-dependent remainder with the exponentially small Bessel sectors before importing fixed- x oscillatory formulas. A second problem is the asymptotic location of the imaginary zeros b_N within their certified pole intervals; their residues are the irregular but explicit divisor weights $\sigma_{-1}(N)$. A third is to exploit the positive Stieltjes representation for rational upper and lower approximants to S , then combine them with the exact gamma-moment remainder. Finally, logarithmic weights in the defining Herglotz sum should yield spectral derivatives of the divisor Dirichlet series, with zeta derivatives and Laurent coefficients entering the transformed residue terms. These are separate extensions; none is presumed by the compact-ratio theorem proved here.

9 Nonseparable reflection jets: a two-generator theorem

The manuscript proves a multivariable product law when the active deformation parameters are separate powers of coordinate variables. Its warning that a general multivariable substitution requires the full Koszul complex is essential. We now give a different sufficient condition: the ideal generated by the active parameters may have arbitrary mixed terms, provided it needs at most two generators. We also exhibit an explicit three-generator obstruction. The results concern the specified formal distribution presentation. They do not assert independence of evaluated polylogarithm or zeta values.

9.1 The distribution presentation and the input theorem

For an integer $q > 2$, put $X_q = (q^{-1}\mathbb{Z})/\mathbb{Z}$. Given weights a_p in a commutative coefficient ring S , one for each prime $p \mid q$, define

$$\mathcal{D}_q(S, a) = \left(\bigoplus_{x \in X_q} S e_x \right) / \left\langle \sum_{\substack{y \in X_q \\ py=x}} e_y - a_p e_x : p \mid q, x \in X_{q/p} \right\rangle. \quad (9.1)$$

Reflection acts by $ce_x = e_{-x}$. Write

$$n = \frac{\varphi(q)}{2}, \quad r = \#\{p > 2 : p \mid q\} + \mathbf{1}_{4 \mid q}.$$

The active sequence is

$$\delta = (a_p - 1 : p > 2, p \mid q), \quad \text{with } a_2(a_2 - 1) \text{ appended if } 4 \mid q. \quad (9.2)$$

There is no active entry at 2 when $v_2(q) = 1$.

We use the following established results from the frozen manuscript [4]. The all-ring basis theorem makes $\mathcal{D}_q(S, a)$ free of rank $\varphi(q)$. In characteristic two, with $N = c - 1$, its reflection homology is

$$\ker N / \operatorname{im} N \simeq \bigoplus_{j=0}^r H_j(K_S(\delta)). \quad (9.3)$$

For a finite free $\mathbb{Z}$ -algebra B , put $\bar{B} = B/2B$ and $C_{\pm} = \mathcal{D}_q(B, a)/(c \mp 1)\mathcal{D}_q(B, a)$. Then

$$\operatorname{Tor}_{\mathbb{Z}} C_+ \simeq \bigoplus_{j \text{ odd}} H_j(K_{\bar{B}}(\bar{\delta})), \quad (9.4)$$

$$\operatorname{Tor}_{\mathbb{Z}} C_- \simeq \bigoplus_{j \text{ even}} H_j(K_{\bar{B}}(\bar{\delta})). \quad (9.5)$$

Every integer-torsion class is killed by two, and both free abelian ranks are $n \operatorname{rank}_{\mathbb{Z}} B$. The source labels are `ir:thm:integral` in `08-integral-02-integral.tex`, `ir:thm:char2-koszul` in `08-integral-05-modular-jets.tex`, and `irjet:thm:finite-free-koszul` in `08-integral-jets.tex`. Thus the algebraic input and the new step below are separately identifiable.

9.2 A colength law without separated parameters

A finite-dimensional commutative k -algebra S is *Frobenius* here if it admits a k -linear functional $\tau : S \rightarrow k$ such that $(u, v) \mapsto \tau(uv)$ is a nondegenerate pairing. The rectangular jet algebra

$$S = k[t_1, \dots, t_d] / (t_1^{L_1}, \dots, t_d^{L_d}), \quad L_i \geq 1, \quad (9.6)$$

is a local Frobenius algebra. Take τ to be the coefficient of $t_1^{L_1-1} \dots t_d^{L_d-1}$: complementary basis monomials pair to one. Its dimension is $T = \prod_i L_i$.

Theorem 9.1 (Two-generator colength law). *Let S be a finite-dimensional commutative local Frobenius algebra over a field k . Let $\mathbf{f} = (f_1, \dots, f_r)$, with $r \geq 1$, and suppose that $I = (f_1, \dots, f_r)$ can be generated by at most two elements. Put $\ell = \dim_k(S/I)$. Then*

$$\dim_k H_j(K_S(\mathbf{f})) = \binom{r}{j} \ell, \quad 0 \leq j \leq r. \quad (9.7)$$

This statement holds in every characteristic.

Proof. If $I = S$, some entry is a unit because S is local. Its two-term Koszul factor is contractible, and both sides of (9.7) vanish. If $I = 0$, the differential is zero and the assertion is immediate.

Otherwise choose a minimal generating subset among the given entries. One obtains it by choosing a basis of $I/\mathfrak{m}I$ from their residue classes and applying Nakayama's lemma. This subset has at most two elements. Permute those elements to the beginning of the list. Every remaining entry is a linear combination of them, so elementary invertible operations change the list into $(f, g, 0, \dots, 0)$, allowing $g = 0$ if only one generator is needed. These operations induce isomorphisms on the free module defining the Koszul complex and on all its exterior powers.

For $r = 1$, the kernel and cokernel of the square linear map $S \xrightarrow{f} S$ have the same dimension, proving the assertion. For $r \geq 2$, first consider the two-entry complex

$$0 \longrightarrow S \longrightarrow S^2 \longrightarrow S \longrightarrow 0.$$

Its zeroth homology is S/I , and its second homology is $\text{Ann}_S(I)$. The Frobenius pairing identifies

$$\text{Ann}_S(I) \simeq \text{Hom}_k(S/I, k) \quad (9.8)$$

as S -modules. Indeed, the orthogonal complement of I is precisely its annihilator. To see the nontrivial inclusion, suppose $\tau(zi) = 0$ for all $i \in I$. For every $a \in S$, one then has $\tau((zi)a) = 0$, because $ia \in I$. Nondegeneracy gives $zi = 0$.

The two end homology groups therefore have dimension ℓ . The Euler characteristic of the complex is $T - 2T + T = 0$, so its middle homology has dimension 2ℓ . Tensoring with the zero-differential exterior algebra on the remaining $r - 2$ generators yields

$$\dim_k H_j = \ell \left[\binom{r-2}{j} + 2 \binom{r-2}{j-1} + \binom{r-2}{j-2} \right] = \ell \binom{r}{j}.$$

No separability of the entries has been used. □

Corollary 9.2 (Nonseparable reflection formula). *Let S be as in Theorem 9.1, now in characteristic two, and put $T = \dim_k S$. If the active ideal generated by (9.2) needs at most two generators, then*

$$\dim_k \mathcal{D}_q(S, a)/(c-1)\mathcal{D}_q(S, a) = nT + 2^{r-1}\ell. \quad (9.9)$$

For $B = \mathbb{Z}[t_1, \dots, t_d]/(t_1^{L_1}, \dots, t_d^{L_d})$, if the reduced active ideal in $S = B/2B$ has that property, both signs have the abelian-group decomposition

$$C_{\pm} \simeq \mathbb{Z}^{nT} \oplus (\mathbb{Z}/2\mathbb{Z})^{2^{r-1}\ell}. \quad (9.10)$$

No splitting as B -modules is asserted.

Proof. Since $N^2 = 0$ in characteristic two, an elementary rank calculation gives

$$\dim_k \text{coker } N = \frac{\dim_k \mathcal{D}_q(S, a) + \dim_k(\ker N / \text{im } N)}{2}.$$

The first dimension is $2nT$. Equations (9.3) and (9.7) make the second $2^r\ell$, proving (9.9). For the integral assertion, both parity sums of the binomial coefficients are 2^{r-1} ; apply (9.4)–(9.5) and the stated free rank. □

In particular, when $r(q) = 2$, this settles the dimension and abelian Smith problem for *every* active sequence over a rectangular jet algebra. The number of deformation variables need not equal the active-prime count. When $r(q) > 2$, the same theorem applies whenever the extra active parameters are redundant generators of the ideal.

9.3 Mixed monomials and a module obstruction

Take $S = k[x, y]/(x^L, y^M)$ and

$$I = (x^a, x^b y^c), \quad 1 \leq b < a \leq L, \quad 1 \leq c \leq M.$$

A quotient basis consists of the monomials $x^i y^j$ with $i < a$ and with either $i < b$ or $j < c$. Consequently

$$\ell = bM + (a - b)c. \quad (9.11)$$

This includes boundary cases where one displayed monomial is zero in S . For two active parameters, the reflected dimension is therefore

$$nLM + 2[bM + (a - b)c].$$

For a concrete integral example, let $q = 15$, $B = \mathbb{Z}[x, y]/(x^3, y^3)$, and take

$$a_3 = 1 + x^2, \quad a_5 = 1 + xy.$$

Here $T = 9$, $r = 2$, $n = 4$, and $\ell = 4$. Thus

$$C_{\pm} \simeq \mathbb{Z}^{36} \oplus (\mathbb{Z}/2\mathbb{Z})^8, \quad \dim_{\mathbb{F}_2} \mathcal{D}_{15}(S, a)/(c - 1) = 44. \quad (9.12)$$

The module statement of the manuscript's separable product law does not extend by replacing its quotient with an arbitrary S/I . In this example, put $Q = S/(x^2, xy)$, over $k = \mathbb{F}_2$. Direct monomial calculation gives

$$\begin{aligned} Q : & \quad 1, x, y, y^2, \\ \text{Ann}_S(I) = (x^2, xy^2) : & \quad x^2, x^2 y, x^2 y^2, xy^2. \end{aligned}$$

The first module is cyclic. The second needs two generators: modulo $\mathfrak{m} \text{Ann}_S(I)$, the classes of x^2 and xy^2 are independent. Thus (9.5) gives

$$\text{Tor}_{\mathbb{Z}} C_- \simeq Q \oplus \text{Ann}_S(I), \quad \text{Tor}_{\mathbb{Z}} C_- \not\simeq Q^2. \quad (9.13)$$

The left-hand module requires three generators and the right-hand candidate requires two. Frobenius duality preserves length; it does not identify an arbitrary finite module with its dual. This explains why the dimension law survives although that stronger module formula fails.

For comparison, one may introduce a redundant third active parameter at $q = 105$ by taking

$$a_3 = 1 + x^2, \quad a_5 = 1 + xy, \quad a_7 = 1 + x^2 y + xy^2$$

over the same ring. The active ideal is unchanged. The homology dimensions become $(4, 12, 12, 4)$, the integral free rank is 216, the torsion multiplicity is 16, and the binary reflected dimension is 232.

9.4 A sharp boundary: three generators can change the answer

Proposition 9.3 (A three-generator obstruction). *Let*

$$S = \mathbb{F}_2[x, y, z]/(x^2, y^2, z^2), \quad \mathbf{f} = (xy, xz, yz).$$

Then $\dim_k S = 8$, $\dim_k S/(\mathbf{f}) = 4$, and

$$(\dim H_0, \dim H_1, \dim H_2, \dim H_3) = (4, 14, 14, 4). \quad (9.14)$$

In particular the total homology dimension is 36, whereas the unsupported colength extrapolation $2^3 \dim_k S/(\mathbf{f})$ would give 32.

Proof. The first differential has image spanned by xy, xz, yz, xyz , and thus rank four. With exterior basis vectors denoted by e_i, e_{ij}, e_{123} , the third differential has image spanned by

$$yze_{12} + xze_{13} + xye_{23}, \quad xye_{12}, \quad xye_{13}, \quad xye_{23}.$$

These vectors are independent, so its rank is also four.

The middle differential has image spanned by

$$\begin{aligned} & xze_1 + xye_2, \quad yze_1 + xye_3, \quad yze_2 + xze_3, \\ & xye_{12}, \quad xye_{23}, \quad xye_{31}. \end{aligned}$$

The first three are independent in degree two, and the last three are independent in degree three. Multiplying any of the first three by x, y, z produces a vector in the span of the final three or zero; higher products vanish. This list therefore spans the image and has rank six. The chain dimensions are 8, 24, 24, 8, so the differential ranks 4, 6, 4 give exactly (9.14). $\square$

At $q = 105$, use the integral algebra $B = \mathbb{Z}[x, y, z]/(x^2, y^2, z^2)$ and weights

$$a_3 = 1 + xy, \quad a_5 = 1 + xz, \quad a_7 = 1 + yz.$$

Here $n = 24$, and the correct reflection dimension is

$$24 \cdot 8 + \frac{4 + 14 + 14 + 4}{2} = 210. \quad (9.15)$$

The naive colength substitution would instead give 208. Both integral Tate parity sums equal 18, hence

$$C_{\pm} \simeq \mathbb{Z}^{192} \oplus (\mathbb{Z}/2\mathbb{Z})^{18} \quad (9.16)$$

as abelian groups. The failed extrapolation would predict only sixteen torsion factors. This is a counterexample to a possible extension, not an error in the present manuscript: its separability restriction explicitly excludes the example.

9.5 Independent checks and the remaining module problem

The accompanying program `verify_nonseparable_jets.py` performs exact binary elimination by two independent constructions. One builds the Koszul matrices from wedge bases. The other expands the original prime rows of (9.1) and the reflection rows in point/monomial coordinates; it imports neither a normal form nor a Koszul calculation. The resulting raw dimensions are

Example	Generators	Relation rank	Quotient dimension
Mixed parameters, $q = 15$	135	91	44
Three-generator obstruction, $q = 105$	840	630	210
Redundant third parameter, $q = 105$	945	713	232

Their unreflected dimensions are 72, 384, 432, as required by the all-ring basis theorem. Omitting reflection rows supplies a corruption control: every reported dimension changes. These checks verify the concrete examples; the preceding proofs establish the theorem for all permitted rings and parameters.

For three active generators, Frobenius duality always reduces the homology dimension profile to (ℓ, h_1, h_1, ℓ) , so the reflection defect is $\ell + h_1$. Computing the syzygy invariant h_1 is the first unresolved step beyond the two-generator law. Further useful questions are to classify ideals for which the binomial colength profile still holds, determine when the positive and negative torsion modules are isomorphic, and characterize the deformations realizable by actual arithmetic spectral jets. Those questions must retain the distinction between module structure, total dimension, and relations among analytic values.

10 Source audit, scope corrections, and integration notes

An audit is useful only when it distinguishes an actual error from a theorem that simply does not answer a new question. The inspected bridge chapters already contain many of the corrections described in earlier incoming reports. We found no additional substantive mathematical error in the specific baseline theorems used as inputs here. The following observations identify the boundaries that must be preserved when integrating this work.

10.1 A verified normalization typo in an external source

There is one concrete bibliographical correction. In the published PDF of Gay–Sebbar [12], printed page 4, the unnumbered display immediately after Eq. (1) and before Eq. (2) omits the factor z between the Lerch transcendent and the polylogarithm. With that paper’s preceding zero-based definition, the corrected display is

$$\mathrm{Li}_s(z) = z\Phi(z, s, 1) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}. \quad (10.1)$$

Indeed multiplication by z and the reindexing $m = n + 1$ prove the identity immediately in $|z| < 1$. The omission was checked in the published PDF, so it is not merely an HTML transcription issue. The inspected ProveIt source already has the correct convention: `chapters/09-zero-geometry.tex`, label `zeros:eq:lambda` and its subsequent $a = 1$ specialization, includes the corresponding factor $1/\rho$. This is an external-source typographical correction, not a required change to that manuscript formula.

10.2 What the present results settle

The final research paragraph of `chapters/09-herglotz-jets.tex` asks for a transition theory in which the argument varies with derivative order [5]. Theorem 6.3 answers that question on every compact positive interval of x/r , with all fixed ratio derivatives and a complete inverse-order expansion. Theorem 6.2 also supplies a finite-order inequality on the full positive axis. Corollary 6.4 identifies the leading limit along every extended limiting ratio. A uniformly accurate expansion that also resolves the exponentially small correction as the ratio approaches zero remains a further problem; the compact-ratio theorem does not assert it.

The multivariate discussion in `chapters/08-integral-jets.tex` correctly retains the full Koszul complex outside the one-variable case [4]. Theorem 9.1 gives a new sufficient condition for a closed dimension law: local Frobenius duality and an active ideal needing at most two generators. In particular, it settles the dimension and abelian Smith factors for every active sequence with two entries over a rectangular jet algebra. It does not classify all multivariate torsion modules.

The three-generator example in Proposition 9.3 is a counterexample to an *unrestricted extrapolation*, not to the manuscript’s properly restricted theorem. Similarly, (9.13) disproves a stronger module substitution that is not needed for the valid dimension formula. These distinctions should remain explicit in the canonical manuscript.

10.3 Normalization and domain statements to preserve

1. **Lerch normalization.** With $\Phi(z, s, a) = \sum_{n \geq 0} z^n (n + a)^{-s}$, one has $\mathrm{Li}_s(z) = z\Phi(z, s, 1)$. Accordingly the exponentially shifted function in (2.1) equals $\mathrm{Li}_s(e^{-t})$ at $a = 1$. Dropping the factor e^{-at} would change the exact finite-part expansion. The Laurent sign convention (1.1) must accompany every formula for generalized Stieltjes constants.

2. **Uniformity and limits.** The coefficients $b_{k,j}$ in Theorem 2.3 retain their dependence on k . Their limiting values from Proposition 2.5 cannot replace them in an arbitrary all-order joint expansion without a rate condition linking k and T . Similarly, a fixed-argument Herglotz asymptotic cannot be substituted at $x = r\lambda$ without a new remainder estimate. The exact mixture provides that estimate here.
3. **Antiderivative constants.** The balanced functions in (5.1) have zero mean and, from the second level onward, matching periodic endpoints. A primitive normalized to vanish at a chosen base point usually differs by a polynomial after repeated integration. Correlation identities for the balanced family cannot be transferred unchanged to that other normalization.
4. **Ordinary integrals versus regularized distributions.** At a nonzero shift, the two Hurwitz endpoint singularities are separate and the domain is $\operatorname{Re} s, \operatorname{Re} t < 1$. At zero shift the additional condition $\operatorname{Re}(s + t) < 1$ is essential. Argument differentiation to the digamma level produces a $1/x$ singularity, so an unqualified ordinary-integral extension would be invalid. A finite-part or distributional convention must be stated and its possible endpoint terms proved.
5. **Which function has the imaginary zeros.** Theorem 7.1 classifies the zeros of $H(y) - \zeta(2)/2$. It does not classify the zeros of H itself. The centering is what produces the odd meromorphic function and the positive Stieltjes-transform argument.
6. **Formal rank versus values.** The rank and torsion results concern the distribution presentation (9.1) over its stated coefficient ring. They do not prove linear or algebraic independence of evaluated polylogarithms, Hurwitz jets, generalized Stieltjes constants, or Gamma values. If an arithmetic specialization inverts two, the reflected two-torsion disappears; the choice of lattice therefore matters.

10.4 Numerical pitfalls and the accompanying checks

Two implementation issues appeared during independent verification and were corrected in the accompanying code. These concern evaluation methods, not counterexamples to the analytic identities.

First, direct differentiation in the order of a generic polylogarithm implementation near an integer can evaluate two large pole terms before their cancellation. An infinitesimal finite-difference step may then lose many digits even at apparently generous working precision. The correlation checker uses a resolved fourth-order symmetric stencil with additional precision, and cross-checks it against independent Hurwitz-jet formulas and split endpoint quadrature. The reusable correlation module uses the convergent odd-zeta expansion instead. The resonance checker pairs the poles analytically and uses native Hurwitz order derivatives for its finite-part comparison.

Second, the high-derivative Herglotz formula multiplies small Hurwitz differences by large powers. Its checker adds guard precision depending on both order and argument before this multiplication. Working precision alone is not an error certificate. Every analytic tail estimate in this article bounds omitted terms in exact arithmetic; a complete implementation with rigorous rounding bounds would additionally require interval or ball arithmetic.

Check	Independent evidence recorded in the package
Resonant cancellation	Six direct Lerch comparisons at 100 decimal working digits; largest absolute residual below 3.4×10^{-85}
Stieltjes finite parts	Orders 0, 1, 2 checked against separately summed Hurwitz derivative tails; largest residual below 1.6×10^{-67}
Shifted Hurwitz products	Split quadrature against the polylogarithmic kernel for real and complex orders; residuals below 6.1×10^{-33}
Log-Gamma correlation	Independent quadrature, Hurwitz jets, polylogarithm jets, and the convergent odd-zeta expansion at three interior shifts
Herglotz transition	Independent Hurwitz evaluations, exact central-moment coefficients, finite Taylor inequalities, and Bessel/primitive comparisons
Formal reflection jets	Exact binary elimination in the original point presentation and in separate Koszul matrices; dimensions 44, 210, 232

The numerical rows are diagnostics. The exact rank rows are finite proofs of their particular matrix statements. The general theorems rely on the proofs in the body of the article. The data files label asymptotic differences separately from identity residuals; a truncated asymptotic formula is not expected to agree to working precision.

10.5 Suggested canonical placement

The source files are modular so that the continuation can be integrated without importing a second preamble. The following placement respects the manuscript's existing organization.

Contribution	Suggested destination
Uniform integer-order resonance and finite parts	Lerch boundary continuation; cross-reference the zeta-jet and Stieltjes sections
Shifted products, convexity, and balanced primitives	Chapter 07, adjacent to the existing integration and log-Gamma calculus
Herglotz transition, profile, and antiderivatives	Chapter 09, after the fixed-argument derivative asymptotic
Two-generator theorem and obstruction	Chapter 08, after the finite-free Koszul and multivariate jet discussion

The accompanying `INTEGRATION.md` gives file-level instructions and the theorem labels used here. The source manifest records the frozen repository revision and the locally reviewed source files. The archive contains the complete article, reproducible code, recorded outputs, and figure source. The research contribution is mathematical prose with independent exact and numerical checks; it is not a Lean formalization.

11 A proposed research programme

The results suggest several precise continuations. The questions below are not assertions that the expected extension must be true. Each includes an initial route and a criterion for what would count as progress.

11.1 Uniform resonance and Stieltjes finite parts

- R1. **Complex approach to the singular argument.** Extend Theorem 2.3 from positive t to complex sectors, with an explicit branch of $\log t$ and a region avoiding the nearest singularities $\pm 2\pi i$. The pole-pairing identity still provides a candidate normal form. The substantive issue is a common majorant and a useful condition on the complex centered scale. A first deliverable is a sector theorem with constants uniform on closed subsectors.
- R2. **The shift boundary $a \downarrow 0$.** Determine a uniform form when a , t , and the spectral displacement approach their singular values together. Subtract the elementary $n = 0$ term before expanding G_a : generalized Stieltjes constants themselves become singular at that boundary. The goal is to identify the additional dimensionless ratio and prove an expansion that matches both the compact a theorem and the isolated first summand.
- R3. **Growing spectral jets.** The present estimates allow every fixed number of order derivatives. Find the maximal growth range of the derivative index for which a quantitative finite-part expansion remains effective. Cauchy bounds on the centered Gamma quotient provide factorial estimates, but optimizing the contour radius against the derivative order and T is necessary. This would turn the fixed-jet Stieltjes identities into a joint large-index asymptotic calculus.
- R4. **Growing transition windows.** Replace the fixed compact set of u in Theorem 2.3 by a window increasing with T . Specify whether the desired error is absolute or relative, since e^{-u} behaves differently in opposite half-planes and the leading profile has nonzero imaginary zeros. The exact expression (2.9) gives a starting point for explicit estimates in terms of $|u|/T$. A useful first result would identify maximal real windows for each fixed truncation order and then treat complex sectors separately.

11.2 Shifted Gamma and polygamma calculus

- G1. **Concavity beyond the first Stieltjes primitive.** For which $r \geq 2$ is the squared increment $E_{r,r}(a)$ concave on $(0, 1)$? The all-order formula (3.12) gives both the cusp and the analytic coefficients, but the first-level coefficient sign proof does not automatically persist. A rigorous counterexample would be as informative as a classification. Compute symbolic coefficient signs and use certified bounds before proposing a universal higher-order conjecture.
- G2. **A distributional polygamma extension.** Choose a specific periodic finite-part normalization for ψ and its argument derivatives, then extend the translated correlation kernel. Determine the delta-function terms at integral shifts and prove that argument differentiation commutes with the chosen normalization. The Fourier formula gives the spectral candidate; the endpoint terms are the essential missing proof.
- G3. **Three-factor correlations and rational shifts.** Develop integrals of three translated Hurwitz jets. Their Fourier coefficients obey a relation $n_1 + n_2 + n_3 = 0$, leading to constrained double sums and a natural comparison with the manuscript's Tornheim layer. At rational shifts, determine explicit character decompositions and minimal formal constant spaces at fixed weight and jet order. Any assertion about independence of the resulting values requires a separate arithmetic argument.
- G4. **Uniform certificates for mixed cusps.** Generalize Corollary 4.4 from $E_{1,1}$ to $E_{r,m}$, with a bound explicit in all three variables a, r, m . A useful target is an error bound that remains practical after reducing the shift to $0 \leq a \leq 1/2$. Such a result would also support interval evaluation of the higher concavity questions.

11.3 The Herglotz transition and its arithmetic profile

- H1. **Matching the small-ratio and fixed-argument regimes.** Combine the exact gamma mixture with the Bessel formula for S to obtain a remainder estimate that stays useful when $\lambda = x/r$ approaches zero with r . The phase of the exponentially small term changes across the gamma distribution. A compact-ratio Taylor bound does not capture that phase at arbitrary shrinking ratios. The desired theorem should recover the existing fixed- x oscillation and the new transition profile from a single controlled representation.
- H2. **Quantitative imaginary-zero geometry.** Theorem 7.1 gives exactly one simple centered zero between successive imaginary poles. Determine the location b_N within $(2\pi N, 2\pi(N+1))$ as $N \rightarrow \infty$, including the effect of the divisor weights. Prove explicit bounds before seeking a limiting fractional position: the residues fluctuate arithmetically. A related goal is a canonical product with justified convergence and normalization.
- H3. **Weighted profiles and higher polylogarithm primitives.** Replace n^{-2} in the positive profile by $n^{-\beta}$ for $\beta > 1$. The same elementary kernel suggests divisor weights of a different order, a changed Mellin product, and an iterated primitive ladder. Determine exactly which properties survive: complete ratio expansions, imaginary-axis interlacing, Bessel representations, and closed normalized polylogarithm antiderivatives. Treat analytic continuation in β separately from the positive-measure range.
- H4. **Rigorous numerical evaluation over the full axis.** Implement ball-arithmetic evaluation that switches between the large-ratio zeta expansion and the small-ratio Bessel expansion using proved total error bounds. For the original finite- r function, combine the central-moment certificate with a direct Hurwitz method. The stopping rule must account for rounding error as well as omitted terms, especially when measuring exponentially small oscillations.

11.4 Formal jets, module structure, and special values

- A1. **Three active generators.** For a local Frobenius algebra, classify ideals for which the homology profile is $(\ell, 3\ell, 3\ell, \ell)$ and those for which the middle dimension is larger or smaller. The explicit example (xy, xz, yz) shows that colength alone is insufficient. A first tractable class is monomial ideals in rectangular jet algebras, where the first syzygies admit a combinatorial presentation.
- A2. **Beyond abelian Smith factors.** Determine the B -module extensions in the reflected integral quotients, including when the two signs have isomorphic torsion modules and when their torsion-free quotients are free over B . The annihilator example (9.13) supplies a small test case. The correct invariants must retain module generators and extension data, not just dimensions after reduction modulo two.
- A3. **Arithmetic realizability of deformations.** Which mixed active sequences arise from actual families of polylogarithm or Hurwitz spectral jets after choosing an integral coefficient lattice? Construct the specialization map explicitly and track which primes are inverted. This separates the abstract Frobenius-algebra theorem from questions about values of analytic functions.
- A4. **The unresolved S_6 special value.** Retain the manuscript's exact definition and current conjecture as the target; the results above do not decide it. A productive next step is to combine the existing integral reductions with a precisely specified space of multiple-polylogarithm periods, prove the functional reductions in that space, and only then test any proposed basis relation. High-precision integer-relation evidence may guide a candidate identity, but cannot supply the missing reduction or independence proof.

The first integration priorities are the two resolved extensions, the strict convexity theorem with

its computable series, and the explicit three-generator obstruction. The next analytic priority is a uniform matching theorem for the Herglotz endpoints; the next algebraic priority is a syzygy description beyond two generators. These are concrete continuations of the proofs given here.

References

- [1] V. Reshetnikov and contributors, *Polylogarithms and their Arithmetic Bridges*, ProveIt repository, canonical manuscript and README, revision 570b0567f311cf1890865065896be2665f469e4f, 10 October 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/570b0567f311cf1890865065896be2665f469e4f/Analysis/Polylogarithms/docs/manuscript>.
- [2] ProveIt contributors, *Lerch boundary continuation*, preserved report, `Analysis/Polylogarithms/docs/reports/lerch-boundary-continuation/article.tex`, same frozen revision as [1]. <https://github.com/VladimirReshetnikov/ProveIt/blob/570b0567f311cf1890865065896be2665f469e4f/Analysis/Polylogarithms/docs/reports/lerch-boundary-continuation/article.tex>.
- [3] ProveIt contributors, Chapter 07, integration and Stieltjes-jet calculus, `chapters/07-integration.tex`, same revision as [1]. <https://github.com/VladimirReshetnikov/ProveIt/blob/570b0567f311cf1890865065896be2665f469e4f/Analysis/Polylogarithms/docs/manuscript/chapters/07-integration.tex>.
- [4] ProveIt contributors, Chapter 08, integral reflection and finite-free jets, `chapters/08-integral-jets.tex` and its input theorems, same revision as [1]. <https://github.com/VladimirReshetnikov/ProveIt/blob/570b0567f311cf1890865065896be2665f469e4f/Analysis/Polylogarithms/docs/manuscript/chapters/08-integral-jets.tex>.
- [5] ProveIt contributors, *Rational Herglotz derivatives and an exponentially small oscillation*, `chapters/09-herglotz-jets.tex`, same revision as [1]. <https://github.com/VladimirReshetnikov/ProveIt/blob/570b0567f311cf1890865065896be2665f469e4f/Analysis/Polylogarithms/docs/manuscript/chapters/09-herglotz-jets.tex>.
- [6] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Section 25.11, Hurwitz zeta function. <https://dlmf.nist.gov/25.11>; accessed 10 October 2026.
- [7] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Section 25.14, Lerch transcendent, especially Eq. 25.14.3_1. <https://dlmf.nist.gov/25.14>; accessed 10 October 2026.
- [8] D. H. Bailey and J. M. Borwein, *Crandall's computation of the incomplete Gamma function and the Hurwitz Zeta function, with applications to Dirichlet L-series*, manuscript dated 15 January 2015, especially Theorem 1 on integer-order polylogarithm derivatives. <https://www.davidhbailey.com/dhbpapers/lerch.pdf>.
- [9] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Sections 25.11 and 25.12. <https://dlmf.nist.gov/25.11> and <https://dlmf.nist.gov/25.12>; accessed 10 October 2026.
- [10] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 1*, Ramanujan J. **6** (2002), 159–188. Preprint: *On some definite integrals involving the Hurwitz zeta function*, <https://arxiv.org/abs/math/0012078>.

- [11] O. Espinosa and V. H. Moll, *A generalized polygamma function*, Integral Transforms Spec. Funct. **15** (2004), 101–115. <https://doi.org/10.1080/10652460310001600573>. Preprint: <https://arxiv.org/abs/math/0305079>.
- [12] R. Gay and A. Sebbar, *Pseudo-differential operators on the circle, Bernoulli polynomials*, Quantum Stud.: Math. Found. **11** (2024), 1–25. <https://doi.org/10.1007/s40509-024-00316-9>.
- [13] D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*, preprint, 2020. <https://arxiv.org/abs/2012.15805>.
- [14] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Section 4.36, infinite products and partial fractions. <https://dlmf.nist.gov/4.36>; accessed 10 October 2026.
- [15] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Sections 10.32 and 10.40, integral representations and large-argument asymptotics of modified Bessel functions. <https://dlmf.nist.gov/10.32> and <https://dlmf.nist.gov/10.40>; accessed 10 October 2026.
- [16] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Section 5.5, functional relations for the Gamma function. <https://dlmf.nist.gov/5.5>; accessed 10 October 2026.
- [17] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Section 25.4, reflection formulas for the Riemann zeta function. <https://dlmf.nist.gov/25.4>; accessed 10 October 2026.